

**Ellipsis, Exponence, and Output Optimization:
A Study of the Phonological ECP**

This paper pushes forward the theories of linearization, allomorph selection, and ellipsis by documenting a series of alternations that emerge before ellipsis sites in Mandar (Austronesian; Central Indonesia). When functional heads are stranded by vp-ellipsis in that language, they are forced to form prosodic words and subjected to top-down phonological constraints at the edges of the phonological phrase and the intonational phrase. These restrictions drive three interlocking repairs: in some contexts, certain heads supplete and other heads move to the left, and in other contexts, prosodically deficient heads that need hosts can force the phonology to suspend deletion of material inside of ellipsis sites. Drawing together the literatures on Match Theory (Selkirk, 2009, 2011) and Distributed Morphology (Halle & Marantz, 1993), I thus argue that linearization, exponence, and ellipsis must ultimately be finalized in the phonology in a process of imperfect translation that allows mismatches to emerge in response to output constraints.

Keywords:

Ellipsis, Exponence, Distributed Morphology, Syntax-Prosody, Match Theory

1 Introduction

Many lines of evidence have converged to suggest that the representations of the syntax are linked to their surface counterparts by a gradual process of translation that passes through two modules on the PF branch. The first of these is the *Morphological Module*, familiar from Distributed Morphology (DM: Halle & Marantz 1993): one which inserts and deletes terminal nodes, reorganizes them through processes of head-movement, fission, and fusion, and parses them into morphological words that provide the domains for other morphological operations and constraints (Embick, 2000; Embick & Noyer, 2001; Arregi & Nevins, 2012; Norris, 2014; Harizanov & Gribanova, 2019; Hewett, 2023). The second is the *Phonology*, which reworks the output of the morphosyntax in the same way (Selkirk, 2009, 2011): reinstantiating its constituent structure in phonological terms (Nespor & Vogel, 1986; McCarthy & Prince, 1993), subjecting it to output-oriented markedness constraints (Prince & Smolensky, 2004) and forcing phonological processes and shifts in constituency as surface repairs (Kubozono, 1989; Selkirk & Elordieta, 2010).

This paper aims to push forward our understanding of how ellipsis operates along this branch. The standard theory takes many types of ellipsis to involve suppression of syntactic structure at PF (Ross, 1969; Hankamer & Sag, 1976; Merchant, 2001). The literature on DM has established that it is distinct from the later processes of reduction and deletion that occur in the phonology, like deaccentuation (Tancredi, 1992; Chomsky & Lasnik, 1993) and surface-level deletion (Napoli, 1982; Booij, 1985; Weir, 2014). As a result, the consensus view treats ellipsis as a type of non-exponence (Wilder, 1997; Bartos, 2000, 2001; Kornfeld & Saab, 2002; Nunes & Zocca, 2009): the constituents marked for ellipsis fail to undergo Vocabulary Insertion, which matches terminal nodes with exponents from the lexicon. But against this backdrop, many questions remain: how does ellipsis interact with the systems of the syntax (Aelbrecht, 2010, 2012;

Gribanova, 2017; Sailor, 2018; Manetta, 2020; Landau, 2020), the operations of the morphology (Saab & Lipták, 2016; Van Craenenbroeck & Temmerman, 2017; Murphy, 2018; Banerjee, 2021), and phonological output constraints (Bennett *et al.*, 2019)?

To refine this picture, this paper lays out a network of interactions that emerge before ellipsis sites in Mandar (Austronesian; Indonesia). These effects are driven by an output constraint that I will term the *Phonological ECP*: a pressure that forces stranded functional heads to form prosodic words. Its effect can be seen in the English paradigm in (1): before ellipsis sites, the unmarked forms of many auxiliaries are ruled out (King, 1970; Bresnan, 1971; Zwicky, 1977; Kaisse, 1983, 1985; Selkirk, 1995; Anderson, 2008a).

(1) *The Phonological ECP*

- a. I'd never go, but [_ω you] [_ω should / could / would / will / have / are]
b. [_ω [_ω you] *sh' *c' **d **ll **ve **re]

Mandar has two types of ellipsis that strand functional heads in this way: the process of VP-ellipsis in (2a), which spares the subject and all auxiliaries in the TP, and the process of Polarity Ellipsis in (2b), which spares a single auxiliary that raises to c⁰.

(2) *Two Types of Ellipsis*

- a. iCicci' na lulus, tapi iKaco' **ndang** na' Δ_{voice}.
NAME FUT graduate but NAME not FUT
'Chichi' will graduate, but Kacho' won't.'
- b. Ma'ua' mua' iKaco' na lulus tapi **ndang** Δ_{TP}.
I said that NAME FUT graduate but not
'I said that Kacho' would graduate but he won't.

The key effects emerge around the future proclitic *na*, which cannot form its own word. Whenever this element should be stranded alone, it forces a salvation effect: the highest head that can form a word in the ellipsis site is spared. In contexts of Polarity Ellipsis, as a result, the positive response to (3a) preserves the lexical verb in (3b).

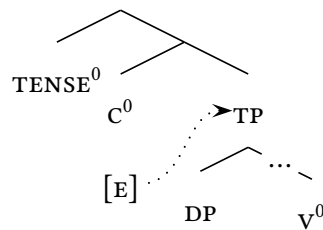
(3) *Temporal Salvation*

- a. **Ndang** bandi inggannanna sola-mu na lulus?
 not VERUM all friend-2GEN FUT graduate
 ‘Won’t all your friends graduate?’
- b. Iye’, **na lulus** Δ.
 yes FUT graduate
 ‘Yes, they will.’

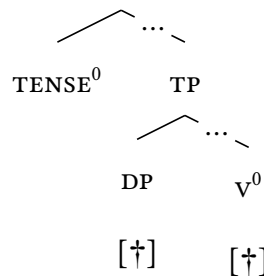
To capture this effect, I propose that Vocabulary Insertion proceeds in two steps: terminal nodes are paired with exponents by MATCH and exponents are inserted by INSERT (see also [Bye & Svenonius 2012](#); [Kalin 2022](#); [Hewett & Kramer 2026](#)). I propose that INSERT occurs in the phonology in the constraint-based calculus where prosodic structure is built ([Selkirk, 2009, 2011](#)). This move licenses an interaction that I will term *Phonological Suspension* (see also [Bennett et al. 2019](#)): when constituents are targeted for ellipsis in the syntax (4a), their terminal nodes are marked with features that block INSERT in the morphology (4b), but when these representations are passed to the phonology, these nodes to be exponed to satisfy output constraints (4c).

(4) *Phonological Suspension*

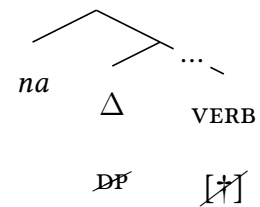
a. *Syntax*



b. *Morphology*



c. *Phonology*

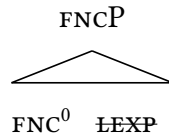


To flesh out the account, I then work out the specific phonological consequences of ellipsis. In Mandar, I show that the Phonological ECP emerges from two constraints (see also [Anderson 2008a](#); [Itô & Mester 2019](#)): the preference to build phonological phrases

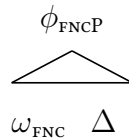
around every XP (Selkirk, 2009, 2011) and the requirement to build prosodic words in every phonological phrase (Nespor & Vogel, 1986). These pressures force stranded heads to form words in the phonological phrases built around their corresponding xps (5a)-(5b). I argue that the future proclitic cannot satisfy this constraint (5c)—and as a result, these pressures interact to force the Suspension Effect in (5d).

(5) *The Structure of the Phonological ECP*

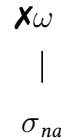
a. *Syntax*



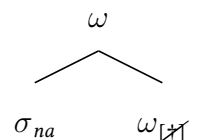
b. *Prosody*



c. *Subcat*



d. *Suspension*



The output constraints that emerge in this context drive two further effects. The first is a type of allomorphy conditioned by top-down prosodic context (Tucker & Henderson, 2010; Henderson, 2012; McCarthy, 2012): at the right edge of the utterance, the negator retains the form *ndang* (6a), but when stranded in other positions, it takes the form *andiang* (6b). The second is a linear shift: when *na* is stranded below the negator in final positions, it remains in place, as in (2a) above, but in non-final positions, it shifts left (6c).

(6) *Suppletion and Movement*

a. iKaco' na lamba, tapi iCicci' **ndang** Δ.
 NAME FUT go but NAME not
 'Kacho' will go, but Chichi' didn't.'

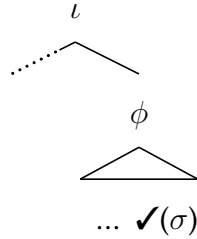
b. iKaco' na lamba, tapi iCicci' **andiang** Δ a?
 NAME FUT go but NAME not PRT
 'Kacho' will go, but Chichi' didn't, eh?'

c. iKaco' na lamba, tapi iCicci' **na** **andiang** — Δ a?
 NAME FUT go but NAME FUT not PRT
 'Kacho' will go, but Chichi' won't, eh?'

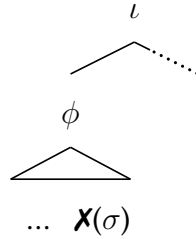
I show that these effects trace a top-down branching constraint (Itô & Mester, 1992; Booij, 1999; Prieto, 2005; Elordieta, 2006): at the right edge of the intonational phrase, the language favors monosyllabic feet (7a), but at the right edge of the phonological phrase, it favors disyllabic feet (7b). At the edges of phonological phrases that are non-final in the intonational phrase, I argue that these constraints guide allomorph selection of the negator (7c), yielding a type of output-optimizing suppletion (Mester, 1994; Kager, 1996; Perlmutter, 1998; Booij, 1998), and relinearize the future proclitic (7d), yielding a type of phonological push movement (Halpern, 1995; Bennett *et al.*, 2016; Kusmer, 2020). I take these effects to suggest two further architectural results: the phonology can rework the linearization of the morphology and it can select among allomorphs when the morphology leaves it with a choice (Tranel, 1996; Rubach & Booij, 2001; Klein, 2003).

(7) *Movement and Suppletion*

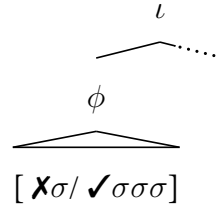
a. *Right Edge*



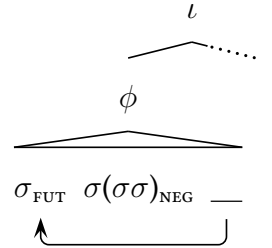
b. *Separation*



c. *Suppletion*



d. *Movement*



The remainder of this paper is structured as follows. Section 2 introduces Mandar, the auxiliaries, the systems of vPE and Polarity Ellipsis, and the behavior of the future proclitic. Section 3 proposes a modification the DM theory of exponence to capture the facts of Temporal Salvation. Section 4 then develops an account of the Phonological ECP and Section 5 argues that it can guide the final stages of allomorph selection and rework the linear order of exponents after their corresponding terminal nodes have been linearized in the morphology. Section 6 concludes.

2 Background

Mandar is spoken by 400,000 people in the Indonesian province of West Sulawesi. It belongs to the South Sulawesi Subfamily, a primary branch of Malayo-Polynesian (Esser, 1938; Mills, 1975), and it is the subject of several descriptive works by the Language Office of South Sulawesi (Pelenkahu *et al.*, 1983; Sikki *et al.*, 1987; Muthalib & Sangi, 1991). This paper describes standard Mandar, which is spoken along the south coast of West Sulawesi (Grimes & Grimes, 1987). It presents data gathered in elicitation with Jupri Talib, a speaker whom I have worked with continually since 2018, that has been cross-checked over regular trips to the field in interaction with the wider community.¹

2.1 The Mandar VP

Mandar is a head-marking and head-initial language that shows a consistent basic word order of vso. It shows an ergative-absolutive system of agreement: transitive verbs agree with the s and second-position clitics agree with the transitive o and intransitive s. The language requires postverbal arguments to surface before locative and temporal adjuncts to the *voiceP* (8), controlling for processes of scrambling and extraposition.²

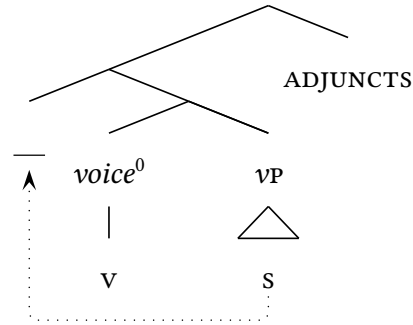
(8) *The Mandar Clause*

U-pissang	i	[_s yau]	[_o iKaco']	[_{CP} mua' pole a']	marondong.
1ERG-tell	3ABS	1SG	NAME	that come 1ABS	tomorrow

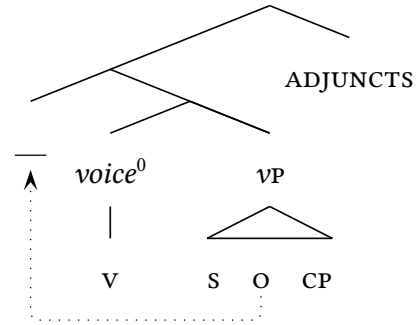
‘I’ll tell Kacho’ that I’ll come tomorrow.’

The vso string typically forms a tight domain that corresponds to the extended vp. The verb raises to a head at the edge of this domain, *voice*⁰, and the absolutive DP undergoes a step of A-movement to the same edge that is typically covert. Arguments and complement clauses are typically realized in their base positions and the adjuncts that follow the vso string are adjoined to the *voiceP*, yielding the vp-level syntax in (9).

(9) a. *The Intransitive VoiceP*



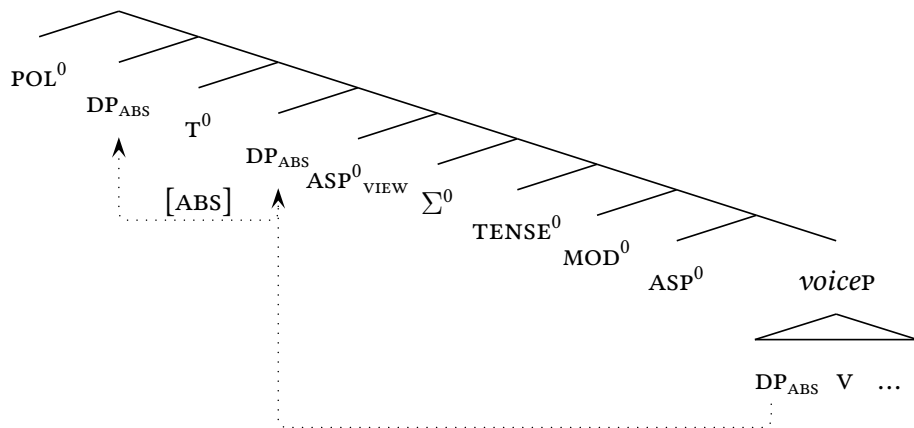
b. *The Transitive VoiceP*



2.2 The Auxiliary System

High in the extended TP, there are two further heads that define subject positions for the absolutive DP. The first of these is viewpoint aspect (Smith, 1991; Demirdache & Uribe-Etxebarria, 2000, 2004), which hosts two aspectual enclitics, and the second is T^0 , which hosts absolutive agreement. These enclitics fall between two types of heads that mark POLARITY (Laka, 1990, 1993; Zanuttini, 1997): a higher head $POLARITY^0$, associated with the expression of *verum focus* (Höhle, 1992; Gutzmann, 2012; Lohnstein, 2016; Gutzmann *et al.*, 2020), and a lower head Σ^0 that hosts sentential negation. Beneath Σ^0 then sit a string of heads that mark tense, modality, and various distinctions of aspect (Travis, 1991; McClure, 1994; Borer, 1994, 1998; Ramchand, 1997; Cinque, 1999, 2013).

(10) *The Mandar TP*



The lower heads in this space host a series of elements that I will term *Auxiliaries*. These elements share a range of properties that establish them as a class: they are rigidly ordered and typically fail to form prosodic words, unlike regular raising verbs, they block verb-raising to Σ^0 , and the highest overt auxiliary raises to Σ^0 and hosts the clitics in $\text{ASP}^0_{\text{VIEW}}$ and T^0 (which otherwise follow the verb). Their inventory is presented in (11).

(11) *The Auxiliary System*

Σ^0		TENSE^0		MOD^0		$\text{ASP}^0_{\text{HAB}} / \text{ASP}^0_{\text{EXP}}$		$\text{ASP}^0_{\text{COMP}}$	
<i>ndang</i> not		<i>pura</i> PAST		<i>mala</i> can/may		<i>byasa</i> usually		<i>yari</i> COMPLETE	
<i>da</i> don't		<i>na</i> FUT		<i>pali</i> have time		<i>rua</i> ever		<i>lesse'</i> INCOMPLETE	

These elements share two properties with the well-established class of auxiliaries in Indonesian (Fortin 2007a,b; Arka 2013; Jeoung 2020; Vander Klok & Paul 2021; see also Vander Klok 2008, 2013a,b, 2015, 2017 on Javanese). First, they fall beneath the highest subject position in the middle field. This pattern can be read from the facts of linear order: there are many configurations that force the absolutive DP to be spelled out in SPEC,TP, rather than its base position, and in that context it precedes the highest AUX. For instance, it is realized in this position when it hosts the quantifier *inggannanna* ‘all’ (12).

(12) *Beneath the Absolutive DP*

- a. $[_{\text{TP}}$ **Inggannanna sola-u** $[_{\text{POLP}}$ *byasa* *i* *yari* *lamba* $]$].
all friend-1GEN usually 3ABS COMPLETE go
‘All my friends usually do go.’
- b. $[_{\text{TP}}$ **Inggannanna tau** $[_{\text{POLP}}$ *pura* *i* *u-bengang* *inggannanna ande* $]$].
all people PAST 3ABS 1ERG-give all food
‘I gave all the people all the food.’

Second, in polar questions the highest auxiliary raises further to a position that precedes the highest subject position and associates with the verum-focus marker *bandi* (13). I take this process to be forced by POLARITY^0 , on a par with the head-movements that occur in contexts of verum focus in languages like Russian (Gribanova 2017).

(13) *Movement to Polarity*

- a. [_{CP} **Pura** bandi [_{TP} ing. sola-mu — yari lamba]]?
 PAST VERUM.3ABS all friend-2GEN COMPLETE go
 ‘Have all your friends really gone?’
- b. [_{CP} **Byasa** bandi [_{TP} ing. sola-mu — rua na-ita]]?
 usually VERUM.3ABS all friend-2GEN ever 3ERG-see
 ‘Have they really usually seen all your friends?’

This process strictly targets the highest auxiliary in the TP. It cannot draw multiple auxiliaries into positions above the absolutive DP (14a). It is also unable to target lexical verbs. In polar questions without auxiliaries, then, *bandi* lowers to follow the verb (14b).

(14) *Restrictions on Movement*

- a. [_{CP} **Pura** — bandi — [_{TP} ingannanna **yari** lao]]?
 PAST VERUM.3ABS all COMPLETE go
 ‘Have all of them really gone?’
- b. [_{CP} — [_{TP} Inggannanna sola-mu **lamba** bandi]]?
 all friend-2GEN go VERUM.3ABS
 ‘Did all your friends really go?’

2.3 VP-Ellipsis

The auxiliaries license a type of ellipsis that suppresses the *voice_P* and spares everything in the extended TP. This process shows all the canonical properties of VP-ellipsis (VPE; Hankamer & Sag 1976; Lobeck 1987, 1995; Johnson 2001; Goldberg 2005): unlike gapping and stripping in Mandarin, it spares auxiliaries, proceeds between speakers and picks up antecedents in the discourse (15a)-(15b), and occurs in embedded clauses and islands. It is also translated naturally into the type of VPE that exists in Indonesian (on which see Fortin 2007a,b and Vander Klok 2016, 2017 on Javanese), which is the language that I have used (alongside Mandarin) to elicit and discuss the examples in this paper.

(15) *Verb-Phrase Ellipsis*

- a. Arere, yari mi mu-tumae?
oh god COMPLETE now.3ABS 2ERG-propose
'Oh god, you did propose to her?'
- b. Iyo', yari mi Δ tadi'.
yes COMPLETE now.3ABS unfortunately
'Yeah, unfortunately I did.'

This process suppresses the lexical verb and the arguments that remain in the *voiceP*.
The vPE site thus cannot be followed by an overt ergative or absolutive DP (16).

(16) *Suppressing the VoiceP*

- a. Yari a' na-uji iKaco' na yari to-o Δ (*iCicci').
COMPLETE 1ABS 3ERG-test NAME and COMPLETE also-2ABS NAME
IDIOMATICALLY: 'I was tested by Kacho' and you were (*by Chichi), too.'
- b. U-sanga pai ndang lao iKaco', tapi pura i Δ (*do asu o)
1ERG-think YET.3ABS not go NAME but PAST 3ABS that dog PRT
'I thought Kacho' hadn't gone, but [*pro* / *that jackass] had.'

It also preserves the absolutive and aspectual enclitics and every auxiliary in the TP.
When auxiliaries in the antecedent disappear before vPE sites, they cannot be interpreted within them (17). In this respect, every Mandar auxiliary behaves like the English *have* (Sag 1976; Lobeck 1987, 1995; Zagana 1988; Johnson 2001; Aelbrecht & Harwood 2015).

(17) *Sparing Higher Auxiliaries*

- a. U-sanga i [TP lesse' makkulia].
1ERG-think 3ABS INCOMPLETE get a degree
'I think he's in the process of getting a degree.'
- b. Tania, [ndang i lesse' Δ / andiang i Δ].
not so not 3ABS INCOMPLETE not 3ABS
'No, he isn't [in the process of getting a degree / getting a degree].

VPE also spares the higher A-positions of the absolutive DP. This fact can be seen when the absolutive DP is non-contrastive but hosts *inggannanna* “all” in the clause that shows VPE (18a). It can also be seen when the absolutive argument is a contrastive DP: in the clauses that show VPE, Mandar allows that argument to contrast with the absolutive DP in the antecedent, and in that context, it is spelled out in SPEC,TP (18b).

(18) *Sparing Subject Positions*

- a. U-sanga i lulus, tapi **ing. sola-u** andiang i Δ
 1ERG-think 3ABS graduate but all friend-1GEN not 3ABS
 ‘I thought they_i had graduated, but all my friends_i didn’t.’
- b. Ing. sola-u u-sanga i lulus, tapi **iKaco’** andiang i Δ
 all friend-1GEN 1ERG-think 3ABS graduate but NAME not 3ABS
 ‘I thought all my friends had graduated, but Kacho’ didn’t.’

When the absolutive DP is contrastive in this way, it can also undergo a higher step of focus movement into the left periphery (21). This step can be detected from a shift in the TP: it suppresses the absolutive clitics in τ^0 . This is a type of Anti-Agreement Effect (Ouhalla, 1993), familiar in this region (Finer, 1997), and we will return to it in Section 5.

(19) *Focus Movement: Anti-Agreement*

- a. Lulus i iCicci’, tapi [_{FOCP} **iKaco’** [_{TP} ____ **ndang** Δ]].
 graduate 3ABS NAME but NAME not
 ‘Chichi’ graduated, but KACHO’ didn’t.’
- b. iCicci’ rua lao, tapi [_{FOCP} **iKaco’** [_{TP} ____ **ndang** **rua** Δ]].
 NAME once go but NAME not ever
 ‘CHICHI’ once went, but KACHO’ never has.’

These effects set up a classical argument for syntactic structure in the ellipsis site (Ross, 1969; Hankamer & Sag, 1976; Sag & Hankamer, 1984; Hornstein, 1994). Mandar allows A-movement from the sites of VPE: the internal argument escapes in passive clauses like (20a) and in regular transitive clauses like (20b). I take these arguments to

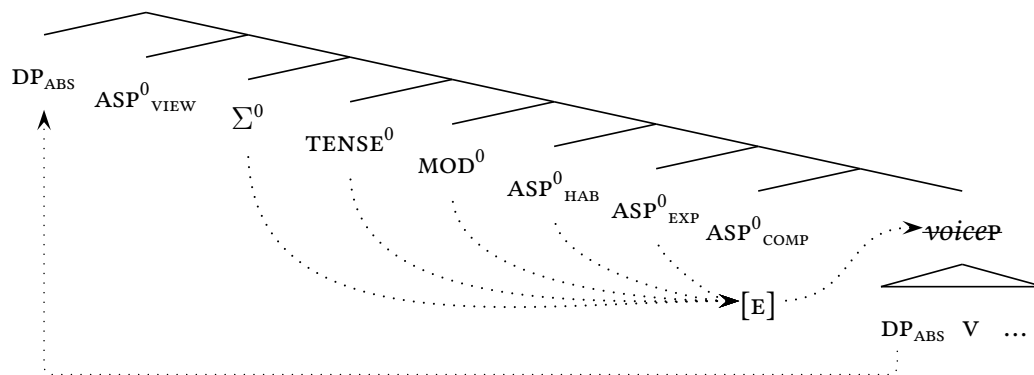
originate in regular internal argument positions in the *voiceP*, on a par with the objects that escape from VPE sites in contexts of Pseudogapping (Jayaseelan, 1990; Lasnik, 1995, 1999; Baltin, 2003; Agbayani & Zoerner, 2004; Johnson, 2009; Gengel, 2013; Thoms, 2016).

(20) *A-Movement from the VPE Site*

- a. Rua i di-ita iCicci', tapi [_{TP} iKaco' ndang i rua Δ].
 once 3ABS PASS-see NAME but NAME not 3ABS ever
 'Chichi' has been seen, but Kacho' hasn't.'
- b. Rua i u-ita iCicci', tapi [_{TP} iKaco' ndang i rua Δ].
 once 3ABS 1ERG-see NAME but NAME not 3ABS ever
 IDIOMATICALLY: 'Chichi' has been seen by me, but Kacho' hasn't.'

To capture these effects, I propose that this process involves PF-suppression of a syntactically-present *voiceP*. Following Merchant 2001, I take it to be driven by a feature [E] that is assigned in the syntax by a selecting head (Van Craenenbroeck & Lipták, 2006; Toosarvandani, 2009; Arregi, 2010; Temmerman, 2012). In that module, I take this feature to be activated through agreement with a higher licensing head (Aelbrecht 2008, 2010, 2012; Landau 2020; see also Lobeck 1987, 1995; Zagona 1988; Johnson 2001): by hypothesis, any of the heads that host auxiliaries in the TP. At this derivational stage, I submit that it has no further effect and allows the absolutive DP to raise from the *voiceP*.

(21) *Ellipsis of the VoiceP*



2.4 Polarity Ellipsis

VPE contrasts with a second process that strands raised auxiliaries in c^0 and suppresses the entire content of the TP: the absolutive and aspectual enclitics, all lower auxiliaries, the verb, and the rest of the *voiceP*. I will refer to this as *Polarity Ellipsis*, as it emerges in contexts that place verum focus on the polarity of the utterance, like the responsive context in (22) (and see also McCloskey 2011, 2017; Lipták 2013; Gribanova 2017).

(22) Polarity Ellipsis

- a. $\begin{array}{c} \downarrow \\ \text{[}_{CP} \text{Yari} \quad \text{bandi} \quad \text{[}_{TP} \text{[ing. sola-mu} \quad \text{]}_{ABS} \text{ — } \text{lao} \quad \text{]} \text{]}? \\ \text{COMPLETE} \quad \text{VERUM.3ABS} \quad \text{all} \quad \text{friend-2GEN} \quad \text{go there} \end{array}$
 ‘Did all your friends really go there?’
- b. Iye’, $\begin{array}{c} \text{[}_{CP} \text{yari} \quad \Delta \quad \text{]} \\ \text{yes} \quad \text{COMPLETE} \end{array}$.
 ‘Yes, [they] did.’

Polarity ellipsis breaks from VPE in three respects. First, it suppresses absolutive agreement, the aspectual enclitics, and all auxiliaries in the TP. In contexts of VPE, absolutive agreement is required unless there is overt focus-movement movement of an absolutive DP (23a) and the aspectual clitics are always able to appear (23b).

(23) Missing Clitics

- a. U-sanga i ndang rua yari lao, tapi rua *(i) yari Δ .
 1ERG-think 3ABS not ever COMPLETE go but once 3ABS COMPLETE
 ‘I thought she had never gone, but she had.’
- b. Minnassa i rua yari lao, tapi yau pa ndang rua Δ .
 clear 3ABS once COMPLETE go but I YET not ever
 ‘It’s clear that they’ve gone, but I haven’t.’

Second, it forces auxiliaries in the TP to be interpreted in the ellipsis site. When auxiliaries are stranded alone, this effect yields a link from the pattern of agreement to

the interpretation of the ellipsis site: when absolutive agreement goes missing without focus-fronting, all remaining auxiliaries in the antecedent must be interpreted in the ellipsis site, and when absolutive agreement is present, they cannot be.

(24) *Interpreting Auxiliaries*

- a. U-sanga i [TP **na** **lesse'** makkulia].
 1ERG-think 3ABS FUT INCOMPLETE get a degree
 'I think he's going to be getting a degree.'
- b. Tania, **ndang** Δ.
 not so not
 'No, [he] isn't [going to be getting a degree].

Third, it suppresses the highest A-position of the absolutive DP. When the absolutive DP hosts *inggannanna* "all," it cannot surface before or after the surviving auxiliary.

(25) *Missing Subjects*

- a. Ing. sola-u u-sanga i na lulus,
 all friend-1GEN 1ERG-think 3ABS FUT graduate
 'I thought all my friends were going to graduate,'
- b. ... **na** (*ing. sola-u) **yari** (*ing. sola-u) Δ.
 and all friend-1GEN COMPLETE all friend-1GEN
 '...and [they / *all my friends] did.'

Fourth, it bans mismatches in the identity of the absolutive DP. The following test reveals this constraint: when contrastive absolutive DPs surface before single auxiliaries and agreement disappears, it is impossible to interpret auxiliaries from the antecedent clause in the ellipsis site (26). This fact shows that these strings are built by VPE.

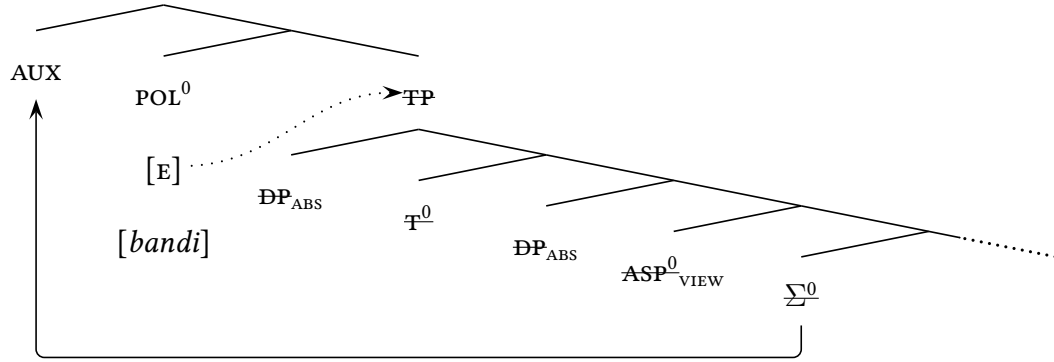
(26) *Matching Requirements*

- a. **Lesse'** bandi lao iKaco'?'
 INCOMPLETE VERUM.3ABS go NAME
 'Is Kacho' in the process of going?'

- b. Tania, inggannanna sola-u **ndang** Δ.
 Not so, all friend-1GEN not
 ‘No, all my friends [are not going / #are not in the process of going].’

To capture these facts, I propose that Polarity Ellipsis proceeds in much the same way as VPE: POLARITY^0 assigns an $[\text{E}]$ feature to its complement, activates it, and attracts the highest auxiliary from Σ^0 to its specifier. This process draws that auxiliary past T^0 and $\text{ASP}_{\text{VIEW}}^0$, allowing the absolutive and aspectual clitics to remain in the TP, and places it above the verum-focus marker *bandi* in POLARITY^0 —yielding a type of head-movement into a specifier position (on which see Harizanov 2019; Harizanov & Gribanova 2019).

(27) *Polarity Ellipsis*



2.5 The Future Proclitic

From this position, we can now turn to a phonologically deviant auxiliary in TENSE^0 : the future proclitic *na*. This element takes the same position as the past-tense *pura*: it follows the negator *ndang* and precedes the modal *mala* and *pali* and the aspectual *byasa*, *rua*, *yari*, and *lesse*’ (28). It never cooccurs with *pura* or the prohibitive *da*.

(28) *The Future Proclitic*

- Ndang i **na** rua yari u-inda’i boyammu!
 not 3ABS FUT ever COMPLETE 1ERG-set foot in your house
 ‘I’ll never set foot in your house!’

The future proclitic breaks from the other auxiliaries in two ways: it cannot host the aspectual and absolutive clitics (29a) and cannot host *bandi* (29b). When it is the highest auxiliary, these elements dock on the next-highest auxiliary (29c) or the verb (29d).

(29) *Problems with Hosting*

- a. [Pura / ***na**] i lao?
 PAST FUT 3ABS go
 ‘They [did / *ll] go.’
- b. [Pura / ***na**] bandi lao?
 PAST FUT VERUM.3ABS go
 ‘[Did / *ll] they go?’
- c. — **Na** yari mi mu-cowa mu-tumae bo’o, de.
 FUT COMPLETE NOW.3ABS 2ERG-try 2ERG-propose again PRT
 ‘Now you’ll try to propose to her again, huh.’
- d. — **Na** mu-cowa bandi?
 FUT 2ERG-try VERUM.3ABS
 ‘Will you try?’

Despite this fact, two patterns suggest that the future proclitic remains independent from lower auxiliaries and verbs in the morphosyntax. First, it prevents other auxiliaries from raising past it to SPEC,POLARITYTP. In the polar questions where other auxiliaries surface beneath it, they follow the absolutive DP (33a), on a par with lexical verbs (33b).

(30) *Intervening for Head-Movement*

- a. — [_{TP} Inggannanna sola-mu **na** yari bandi lamba]]?
 all friend-2GEN FUT COMPLETE VERUM.3ABS go
 ‘Will all your friends really go?’
- b. — [_{TP} Inggannanna sola-mu **na** lamba bandi]]?
 all friend-2GEN FUT go VERUM.3ABS
 ‘Will all your friends really go?’

Second, the future proclitic is split from the verb by VPE when it can lean phonologically on the negator (31a). At the same time, it cannot be carried up to SPEC,POLARITYTP with that head (31b) and cannot be spared beneath it by Polarity Ellipsis (31c). These facts suggest that it is not amalgamated with it by M-merger (Matushansky, 2006), morphological head-movement (Adger *et al.*, 2009; Harley, 2013; Harizanov & Gribanova, 2019), or the mechanism that puts English auxiliaries together with *n't*.

(31) *Stranding under VPE*

- a. iCicci' na lamba, tapi iKACO' **ndang** **na'** Δ.
 NAME FUT go but NAME not FUT
 'CHICHI' will go, but KACHO' won't.'
- b. [_{CP} Ndang — bandi [_{TP} ing. sola-mu **na** lao]]?
 not VERUM.3ABS all friend-2GEN FUT go
 'Won't all your friends really go?'
- c. Tania, ndang (***na**) Δ.
 not so not FUT
 'No, [they] won't.'

I thus propose that the future proclitic deviates from the other auxiliaries in the phonology alone. On this analysis, its special behavior emerges from an output-oriented selectional restriction linked to the exponent *na* (Inkelas 1989; Booij & Lieber 1993; Peperkamp 1997; Raffelsiefen 1999; Bennett *et al.* 2018; see also Rolle & Hyman 2018; Kalin & Rolle 2023). This is the vertical subcategorization frame in (32a), which prevents *na* from forming its own prosodic word (32b). This pressure drives parses like (32c)-(32d): *na* falls into the words that are built around verbs and stranded auxiliaries (Section 5).

- (32) a. *Selection* b. *Constraint* c. *Verbs* d. *Negation*
- ω

na —

ω

X(na)

ω

 σ_{na} ω_{VERB}

ω

 ω_{NEG} σ_{na}

2.6 Temporal Salvation

For reasons laid out in Section 4, this phonological restriction yields a second constraint: the future proclitic cannot be stranded alone by Polarity Ellipsis (33a) or VPE (33b).

(33) Problems with Ellipsis

- a. Ma'ua i mua' na lao i, tapi [ndang / *na] Δ.
 say 3ABS that FUT go 3ABS but NOT / FUT
 'They said that they'd go, but they [WON'T / *LL].'
- b. Lao i iCicci' na iKaco' [pura / *na] to-i Δ.
 go 3ABS NAME and NAME PAST / FUT also-3ABS
 'Chichi went and Kacho' [did / *ll] too.'

When the future proclitic should be stranded by Polarity Ellipsis, that process spares the next-highest head that can form a prosodic word in the TP. I will refer to this effect as *Temporal Salvation*. It can rescue lower auxiliaries (34a) and the lexical verb (34b).

(34) Temporal Salvation

- a. U-sanga i ndang na yari pole, tapi [CP na yari Δ].
 1ERG-think 3ABS not FUT COMPLETE come but FUT COMPLETE
 'I thought they wouldn't come, but [they] WILL.'
- b. U-sanga i ndang na pole, tapi [CP na pole Δ].
 1ERG-think 3ABS not FUT come but FUT come
 'I thought they wouldn't come, but [they] WILL.'

When the future proclitic should be stranded alone by VPE, the same repair rescues the verb (35a). This is impossible when *na* can phonologically lean on the negator (35b).

(35) Rescuing the Verb

- a. YAU ndang makkiring-oto tapi [TP iKaco' na makkiring i Δ].
 I not send-car but NAME FUT send 3ABS
 'I didn't send things by car, but Kacho' will [send things by car].'
- b. YAU makkiring-oto tapi [TP iKACO' ndang na (*makkiring) Δ].
 I send-car but NAME not FUT send
 INTENDED READING: I sent things by car but Kacho' won't [send things by car].'

The configurations that rescue auxiliaries from the site of Polarity Ellipsis contrast with vPE in three respects. First, they require *na* to be the highest auxiliary (36b). Second, they only allow one more auxiliary to appear, they force all other auxiliaries in the antecedent to be interpreted in the ellipsis site, and they they lack absolutive agreement even though they show no focus movement of the absolutive DP (36c).

(36) *Rescuing Auxiliaries*

- a. **Na rua** bandi lesse' makkulia do wattu o?
 FUT ever VERUM.3ABS INCOMPLETE get a degree that time PRT
 'Will she have begun to get a degree by then?'
- b. Tania, [_{CP} ndang (***na** ***rua**) Δ].
 not so not FUT EVER
 'No, [she] WON'T [have begun to get a degree by then].
- c. Iye', [_{CP} **na** rua Δ].
 yes FUT ever
 'Yes, [she] WILL [have begun to get a degree by then].

Third, they impose a matching constraint: the rescued auxiliary must be identical to the highest auxiliary in the antecedent TP (37a). This effect is never forced by vPE (37b).

(37) *The Matching Requirement*

- a. U-sanga i **na yari** lao na na [**yari** / ***rua**] Δ.
 1ERG-think 3ABS FUT COMPLETE go and FUT COMPLETE once
 'I thought he would go and he [WILL / *WILL HAVE].
- b. iCicci' lao na iKaco' na [**yari** / **rua**] to-i Δ].
 NAME go and NAME FUT COMPLETE once also-3ABS
 'Chichi' went and Kacho' [will / will have] too.'

When this process rescues the verb, it shows the same profile: it allows the verb to surface without absolutive agreement in contexts of Polarity Ellipsis and it enforces a matching constraint on the verb in contexts of Polarity Ellipsis (38a) and vPE (38b).

(38) *The Verbal Identity Requirement*

- a. U-sanga i na massapeda dai' na na [**massapeda** / ***lamba**] Δ.
 1ERG-think 3ABS FUT bike up and FUT bike go
 INTENDED READING: 'I thought he would bike up and he WILL.'
- b. iCicci' mi'oro dai' na iKaco' **na** [**mi'oro** / ***mi'enge**] to-i Δ.
 NAME sit up and NAME FUT sit sit also-3ABS
 INTENDED READING: 'Chichi' sat up and Kacho' will too.'

In this context, the following test reveals ellipsis of the *voiceP*. Mandar has several types of constituents that cannot elide on their own, like pseudo-incorporated manner adjuncts and PPs. In AUX-AGREEMENT-VERB strings which cannot possibly involve *VPE*, these elements cannot be interpreted as present when they appear in preceding *VPS* (39).

(39) *Unelidable Constituents*

- a. Byasa i makkiring-**oto sau** na byasa to-a' makkiring.
 usually 3ABS send-car over and usually also-1ABS send
 LITERALLY: 'she usually sends things over by car and I usually send things too.'
- b. *Na-cowa i na-kui'-**lima sung** na yari i na-kui'.
 3ERG-try 3ABS 3ERG-pry-hand out and COMPLETE 3ABS 3ERG-pry
 LITERALLY: 'she tried to pry it out by hand and she pried it.'

In NA-VERB-AGREEMENT strings, the language allows two readings: manner adjuncts and PPs in preceding *VPS* can be interpreted as systematically absent or systematically present (40). To capture this second parse, I propose verb-stranding ellipsis of the *voiceP*.

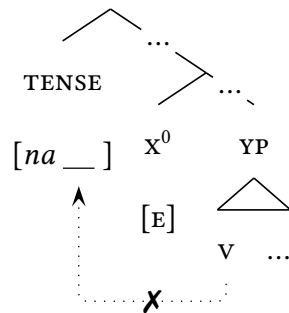
(40) *Verb-Stranding Ellipsis*

- a. Pura i makkiring-**oto sau** na na makkiring to-a' Δ.
 PAST 3ABS send-car over and FUT send also-1ABS
 READING A: 'she sent things over by car and I'll send things too.'
 READING B: 'she sent things over by car and I will [send things over by car] too.'
- b. Melo' i na-kui'-**lima sung** na na na-kui' i Δ.
 want 3ABS 3ERG-pry-hand out and FUT 3ERG-pry 3ABS
 'She wanted to pry it out by hand and she will [pry it out by hand].'

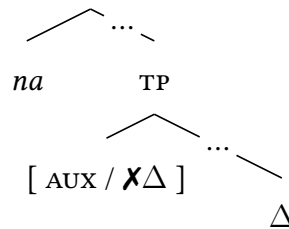
3 Phonological Suspension

Taking stock, then, the future proclitic spells out a head that can license VPE, can escape Polarity Ellipsis, and cannot attract lower heads in the syntax or morphology (41a). In the phonology, however, the prosodic requirements of its exponent drive a Suspension Effect: whenever *na* should fall before an ellipsis site alone, the Phonological ECP prevents suppression of the highest head in the ellipsis site that can form a licit word—sparing verbs and auxiliaries under Polarity Ellipsis (41b) and verbs under VPE (41c).

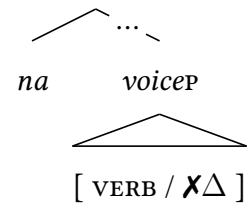
(41) a. *Morphosyntax*



b. *Aux Salvation*



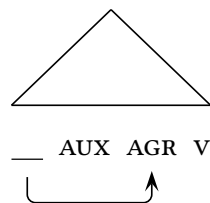
c. *Verb Salvation*



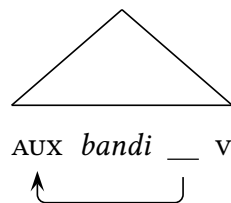
This effect is matched by two more interactions that remove *na* from positions that would force it to form a word: unlike regular auxiliaries which can host second-position elements like absolutive agreement (42a) and *bandi* (42b), *na* is skipped by the system of second-position linearization (42c) and spelled out low when it raises to SPEC,POLP (42d).

(42) *The Wider Conspiracy*

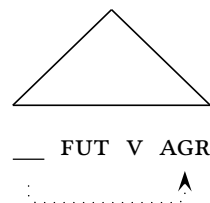
a.



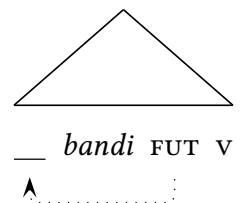
b.



c.



d.



The task of this section is to integrate the Salvation Effect with the standard theory of ellipsis in DM: one that links it to a failure of Vocabulary Insertion (Wilder, 1997; Bartos, 2000, 2001; Kornfeld & Saab, 2002; Nunes & Zocca, 2009; Aelbrecht, 2010, 2012; Temmerman, 2012; Arregi & Nevins, 2014; Merchant, 2015; Abels, 2019; Mendes & Nevins, 2022; Sailor, 2022). I thus propose that the final stage of this process must occur in the phonology, where it can be shaped and ultimately forced by the Phonological ECP.

3.1 Setting the Stage

Ellipsis is distinct from two other types of suppression that operate along the PF branch. The first are morphological processes of structure-removal like Fusion, Pruning, and Obliteration, which target terminal nodes in response to morphological constraints (Embick & Noyer, 2001; Arregi & Nevins, 2012). The second are phonological processes of suppression (Thrasher Jr, 1974; Napoli, 1982; Wilder, 1997; Weir, 2014), like *Pronoun zap* (Ross 1982), coordination reduction (Booij, 1985) and in-situ ellipsis (Abe, 2015; Ott & Struckmeier, 2016, 2018; Kimura & Narita, 2021; Dikken *et al.*, 2000), and the left-edge deletion in (50). At least seven properties separate these processes from ellipsis: they do not require syntactic licensing, do not require antecedents, only target phonologically deaccented constituents, need not target syntactic constituents, can target purely phonological constituents, can be triggered by output constraints, and can seem to retain segmental information from the strings they delete. Zwicky & Pullum 1983, for instance, note that auxiliaries stranded by left-edge deletion can assimilate in voice to their hosts: *is* surfaces as [s] after zapped *it* (43a) and as [z] after zapped *there* (43b).

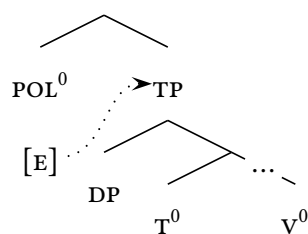
(43) *Phonological Deletion*

- a. Δ [s] really cold out there.
- b. Δ [z] really no more hope.

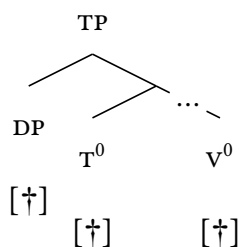
I assume that the terminal nodes in ellipsis sites host features that typically block Vocabulary Insertion (Müller, 2011; Saab, 2008, 2010, 2022; Stigliano, 2022b,a). Following (Müller, 2011), I represent this as the dagger [$\dagger$]. Beneath heads that bear [E], I assume that it is inserted on their complements in the morphology and then distributed to all the terminal nodes that they contain (see also Saab 2008, 2010, 2022; Merchant 2015; on analogous theories of feature percolation in contexts of concord, see Norris 2014). In the TP in (44), then, the nodes that receive [$\dagger$] usually go unexponed when insertion occurs.

(44) *Blocking Insertion*

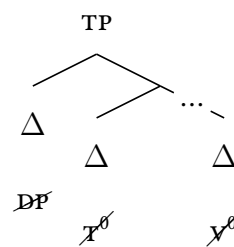
a. *Syntax*



b. *Morphology*



c. *Insertion*



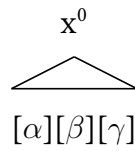
3.2 Vocabulary Insertion

I assume that morphological exponent selection proceeds in a cyclic and bottom-up process that targets terminal nodes in a series of steps. Following Arregi & Nevins 2012, I take this process to follow impoverishment (Halle & Marantz, 1993), node insertion and deletion (Choi & Harley, 2019), fission and fusion (Hewett, 2023), and linearization (Adger 2006; Merchant & Pavlou 2017; Ostrove 2018; see also Embick & Noyer 2001). I assume that it proceeds in three steps in the morphological module that I will refer to collectively as the operation MATCH. First, it identifies the complete set of exponents that can be matched with the target head on the basis of its diacritic identity (Harley, 2005, 2014, 2015; Acquaviva, 2009). Second, it narrows down the set of possible exponents on the basis of the subset principle (Halle, 1997), which favors the candidates

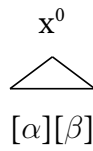
that most closely match the featural specification and structural position of the target (Bobaljik, 2000, 2012; Embick, 2010; Gouskova & Bobaljik, 2020). Third, when the subset principle favors multiple candidates to the same degree, it narrows the set further on the grounds of morphological markedness constraints (Arregi & Nevins, 2012).

(45) *Stage One: Match*

a. *Possibilities*



b. *Subset Principle*



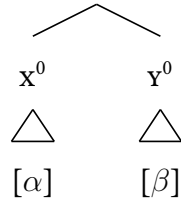
c. *Markedness*



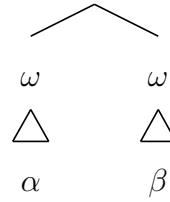
I propose that exponents are inserted into terminal nodes by a second process, INSERT, which occurs in the phonology. I take this second module to operate in the manner familiar from the mainstream literature on the syntax-prosody interface (Inkelas & Zec, 1995; Truckenbrodt, 2007; Bennett & Elfner, 2019), drawing especially from the paradigm of Match Theory (Selkirk 2009, 2011; Elfner 2012, 2015; Elordieta 2015; Bennett *et al.* 2016; Kalivoda 2018; Van Handel 2021; Brodtkin 2025; see also Ishihara & Kalivoda 2022). On this view, all phonological evaluation proceeds in the type of parallel, constraint-based, and optimizing calculus of standard OT (Prince & Smolensky, 2004). In that context, insertion applies in parallel with the translation of morphosyntactic constituents into prosodic domains (Section 4) and the translation of the linear order of morphological nodes to the linear order of their phonological exponents (Section 5). In the simplest cases, these systems translate linearized and unexponed morphological representations like (46a) to linearized and exponed prosodic representations like (46b).

(46) *Stage Two: Insert*

a. *Morphology*



b. *Phonology*



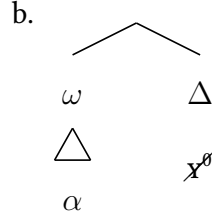
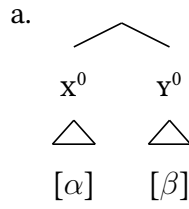
I propose that INSERT shares two properties with the other systems that translate morphological representations into phonological terms. First, it is guided by ranked and violable constraints, including two pressures from the families MAX and DEP. The first forces terminal nodes to undergo INSERT (47a). The second prevents terminal nodes from undergoing INSERT when they host $[\dagger]$ (47b). On this account, the effect of $[\dagger]$ is imposed by a ranked and violable constraint (cf. Müller 2011; Merchant 2015; Saab 2022).

(47) *Constraints on Insertion*

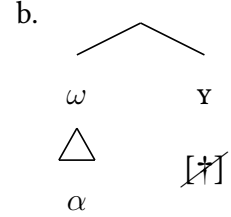
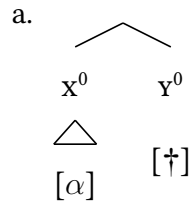
- a. MAX: AOV for every terminal node that fails to undergo INSERT.
- b. DEP: AOV for every terminal node marked $[\dagger]$ that undergoes INSERT.

Second, INSERT interacts with independent phonological constraints. I assume that these pressures can outrank MAX and prevent terminal nodes from being exponed (48a)-(48b), yielding the types of phonologically suspended exponence discussed by Raffelsiefen 1996, 1999, 2004; Kurisu 2001; Henderson 2012, and Aissen 2017. I also assume that they can outrank DEP to force the exponence of constituents marked for ellipsis (49a)-(49b), on a par with the cases where output constraints derail chain reduction and force multiple-copy spell out (Landau, 2006; Kandybowicz, 2006, 2015; Vicente, 2007; Joutteau, 2012; Cheng & Vicente, 2013; Hein, 2017; Bjorkman, 2020).

(48) *Phonological Suppression*



(49) *Phonological Salvation*



3.3 Phonological Suspension

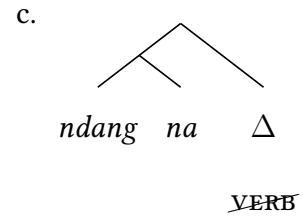
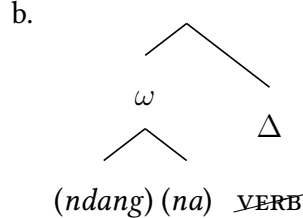
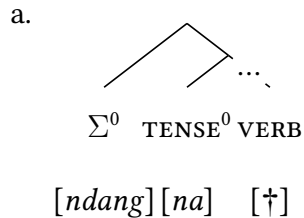
I propose that Temporal Salvation occurs when DEP is ranked beneath two specific constraints. The first is SUBCAT, which enforces the phonological requirements of individual exponents like *na* (Werle, 2009; Rolle & Hyman, 2018; Bennett *et al.*, 2018; Tyler, 2019). The second is the restriction that I term the PHONOLOGICAL ECP, which forces the exponents of stranded functional heads to form prosodic words (Section 4).

(50) *Preparing for Salvation*

- a. SUBCAT: AOV for every subcategorization frame that is not met.
- b. P-ECP: “Stranded functional heads must form prosodic words.”

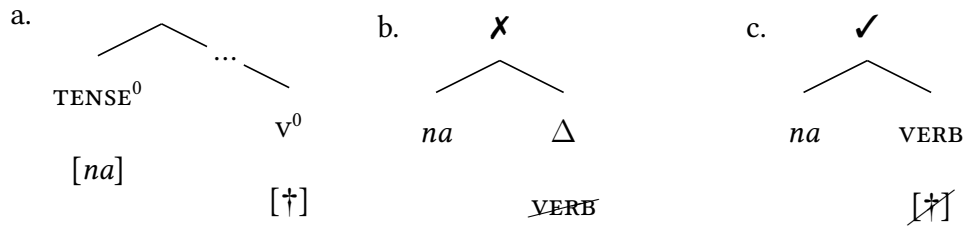
When the future proclitic should be stranded below negation, these constraints force the result in (51a)-(51b): *na* forms a word with the negator and ellipsis targets the complete *voiceP*. This result satisfies DEP, as every head marked $[\dagger]$ goes unexponed, and it improves on alternatives that violate SUBCAT and the Phonological ECP (51c).

(51) *Stranding below Negation*



When the future proclitic should be stranded by ellipsis alone, these constraints force the translation in (52): the Phonological ECP and SUBCAT outrank DEP and INSERT is forced to target the highest head bearing $[\dagger]$ that can form a licit prosodic word.

(52) *The Salvation Effect*



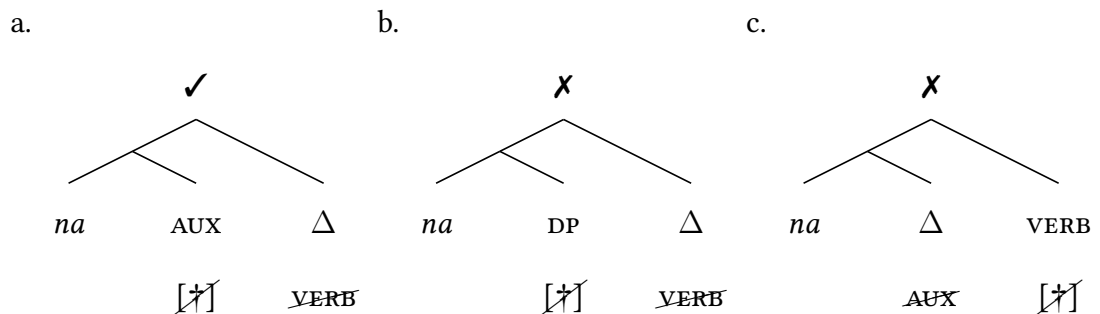
I propose that the preference to spare the highest head emerges from the interaction of two further constraints. The first is a ranked and violable pressure for isomorphism that favors the construction of prosodic domains around constituents that are related in the morphosyntax (53a). The second is a ranked and violable pressure to preserve the relationships of adjacency established by the linearization of the morphology (53b).

(53) *Additional Constraints*

- a. DEP- ω : AOV for every prosodic word that contains exponents of nodes that belong to different extended projections in the syntax.
- b. IDENT-LIN: AOV for every pair of nodes adjacent in the morphology whose exponents are not adjacent in the phonology.

These constraints rule out two alternative strategies to provide the future proclitic with a phonological host: one that would rescue the absolutive DP from Polarity Ellipsis and one that would rescue any head lower than the highest possible auxiliary or verb.

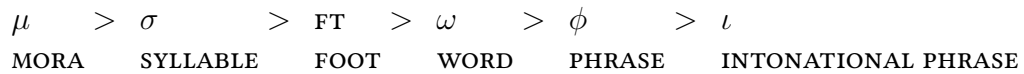
(54) *Impossible Repairs*



4 The Phonological ECP

From this position, we can pin down the phonological pressure that drive these effects. The case begins from the building blocks of hierarchical phonological organization: the universal, abstract, and weakly layered domains that make up the Prosodic Hierarchy (Selkirk 1984, 1986; Nespor & Vogel 1986; Itô & Mester 2007, 2009, 2012, 2013). Three of these domains are important to us: the intonational phrase, which is built around complete utterances, the phonological phrase, which is built around syntactic xps, and the prosodic word, which is built around lexical heads (on the first: Myrberg 2010, 2013; Hamlaoui & Szendrői 2015, 2017; on the second, Truckenbrodt 1995, 1999; Elfner 2012, 2015; Elordieta 2015; on the third, McCarthy & Prince 1993; Selkirk 1995)

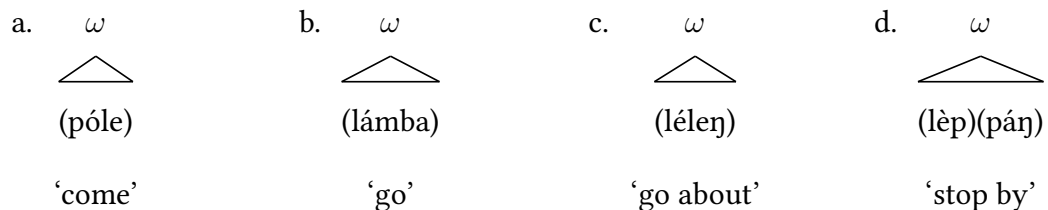
(55) *The Prosodic Hierarchy*



4.1 Identifying Prosodic Domains

In Mandar, these domains can be identified in a regular and straightforward way: their edges impose constraints and host prosodic events that render them visible at the surface (Nespor & Vogel 1986). The prosodic word, to begin, can be identified from the facts of stress: In each word, moraic trochees are built iteratively from the right around pairs of light syllables (56a), heavy-light sequences (56b), and light-heavy sequences (56c).

(56) *The Prosodic Word*



The phonological phrase, in turn, can be identified from the distribution of a final high tone. This tone falls after strings of functional and lexical heads (57a), lexical heads and head-level adjuncts (57b), norman genitives (57c), and names (57d).

(57) *The Phonological Phrase*

- | | |
|---|---|
| <p>a. {_φ tama ri wójaŋ }^H
 tama di bodʒaŋ
 into P house
 ‘Into the house’</p> | <p>b. {_φ dòttór malingáo? }^H
 dottor malingao
 doctor tall
 ‘Tall doctor’</p> |
| <p>c. {_φ bìrit tànŋalálaŋ }^H
 biriŋ taŋŋalalaŋ
 edge road
 ‘Edge of the road’</p> | <p>d. {_φ dʒúpe táli? }^H
 dʒupe tali?
 NAME NAME
 ‘Jupri Talib’</p> |

The intonational phrase, at last, can be detected (among many other ways) from a process of ?-epenthesis that occurs at its edge. In the simplest cases, this process applies after vowel-final words whenever it would license an exhaustive parse: targeting words that contain odd numbers of light syllables (58a), for instance, which would otherwise receive the incomplete parse in (58b). It never applies after words that contain even numbers of light syllables, which are exhaustively parsed in the default case (58c)-(58d).

(58) *The Intonational Phrase*

- | | |
|---|---|
| <p>a. {_ι démmo^H làsuná? }^H.
 diaŋ mo lasuna
 there is now shallot
 ‘Now there are shallots.’</p> | <p>b. {_ι démmo^H lasúna^H á? }^H.
 diaŋ mo lasuna a
 there is now shallot PRT
 ‘Now there are shallots, eh.’</p> |
| <p>c. {_ι démmo^H lémo^H }^H.
 diaŋ mo lemo
 there is now citrus
 ‘Now there are lemons.’</p> | <p>d. {_ι démmo^H tálayáe^H }^H.
 diaŋ mo talagae
 there is now tomato
 ‘Now there are tomatoes.’</p> |

4.2 Parsing Functional Heads

These diagnostics reveal that the auxiliaries typically fail to form independent prosodic words. In Mandar, freestanding functional heads—like auxiliaries, complementizers, prepositions, demonstratives, and the absolutive agreement and aspectual clitics—never host word-level stress (59a)-(59b). This property sets them apart from lexical roots of all types, like nouns, verbs, adjectives, and the middle-field adverbials in (59c)-(59d).

(59) Word-Level Stress

- | | |
|--|--|
| <p>a. na póle^H pe tik kátʃoʔ^H.
 na pole pai itij katʃoʔ
 will come YET.3ABS that NAME
 ‘That Kacho’ will come yet.’</p> | <p>b. itik kátʃoʔ^H mala na póle^H.
 itik katʃoʔ mala na pole
 that NAME can FUT come
 ‘That Kacho’ might’ll come.’</p> |
| <p>c. mé lámba^H mi náwuŋ^H.
 mane lamba mi nauŋ
 just go NOW.3ABS down
 ‘She just went down.’</p> | <p>d. táppa tàndáʔ^H mi .íni^H.
 tappaʔ tandaʔ mi dini
 suddenly arrive NOW.3ABS here
 ‘She suddenly showed up here.’</p> |

They also show that the auxiliaries are typically excluded from the domain of word-level stress, which I take to be the *Minimal Prosodic Word*. The future proclitic and the auxiliaries are never footed together with the initial syllables of following lexical heads (60a)-(60b). They also have no effect on glottal epenthesis, which targets trisyllabic roots and not disyllabic ones even when they appear (60c)-(60d).³

(60) The Stress Domain

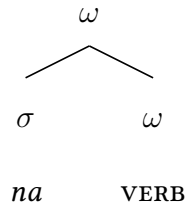
- | | |
|--|---|
| <p>a. i hadára^H na balíli^H áʔ^H?
 i hadara na balili a
 D NAME FUT return PRT
 ‘Hadara will come back, eh?’</p> | <p>b. i rípe^H ra malái^H áʔ^H?
 i ripaʔi rua malai a
 D NAME once go home PRT
 ‘Ripa’i once went home, eh?’</p> |
|--|---|

c. i hadára^H **na** bàlilí?^H.
 i hadara na balili
 D NAME FUT return
 ‘Hadara will come back.’

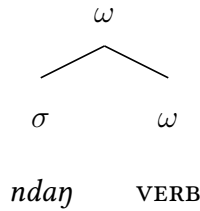
d. i rípe^H **ra** póle^H.
 i ripaʔi rua pole
 D NAME once come
 ‘Ripa’i once came.’

I take these effects to suggest the recursive parse in (61): functional heads typically fall into higher prosodic words with adjacent lexical heads. This proposal traces two well-established results: word-level recursion is possible (Kabak & Revithiadou, 2009; Vigário, 2010; Padgett, 2012; Bennett, 2018) and routinely deployed to parse FNC-LEX strings (Booij, 1996; Peperkamp, 1997; Vigário, 1999; Vigário, 2003; Itô & Mester, 2009).

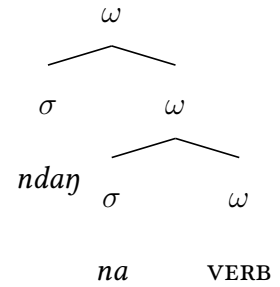
(61) a. *The Future*



b. *Auxiliaries*



c. *Recursion*



4.3 Prosodic Promotion Effects

Immediately before ellipsis sites, the auxiliaries show a phonological shift: they begin to carry word-level stress. This effect typically targets the rightmost auxiliary: the sole element stranded by Polarity Ellipsis (62a) and the rightmost stranded by VPE (62b).

(62) *Prosodic Promotion*

a. usáŋe^H jari lúlus^H na **jári^H** .
 u-saŋa i iari lulus na iari Δ
 1ERG-think 3ABS COMPLETE graduate and COMPLETE
 ‘I thought she graduated and [she] DID.’

b. itʃittʃi^H lúlus^H, tapi ikátʃoʔ^H **ndar ra** **jári^H**
 itʃittʃiʔ lulus tapi ikátʃoʔ ndaŋ rua iari Δ
 NAME graduate but NAME not ever COMPLETE
 ‘CHICHI’ graduated, but KACHO’ didn’t.’

When the rightmost auxiliary is *na*, this requirement must be passed off to one of two targets: the preceding negator, in contexts of VPE (63a), or a following auxiliary or verb, when those elements are rescued from Polarity Ellipsis or VPE (63b).

(63) *Passing off the Pressure*

- a. $\widehat{it}\widehat{fitt}\widehat{ji}^H$ $\widehat{lulus}^H$, $\widehat{tapi}$ $\widehat{ikat}\widehat{fo}^H$ **$\widehat{ndan} \widehat{na}^H$**
 $\widehat{it}\widehat{fitt}\widehat{ji}?$ $\widehat{lulus}$ $\widehat{tapi}$ $\widehat{ikat}\widehat{fo}?$ $\widehat{ndan}$ $\widehat{na}$ Δ
 NAME graduate but NAME not will
 ‘CHICHI’ graduated, but KACHO’ won’t.’
- b. $\widehat{usaje}^H$ $\widehat{na}$ $\widehat{jari}$ $\widehat{lulus}^H$ $\widehat{na}$ **$\widehat{na} \widehat{jari}^H$** .
 $\widehat{u-saja}$ $\widehat{i}$ $\widehat{na}$ $\widehat{iari}$ $\widehat{lulus}$ $\widehat{na}$ $\widehat{na}$ $\widehat{iari}$ Δ
 1ERG-think 3ABS FUT COMPLETE graduate and FUT COMPLETE
 ‘I thought she would graduate and [she] WILL.’

Before VPE sites, English imposes three analogous constraints. First, auxiliaries must form prosodic words when they surface alone (64a)-(64b). Second, they can enclitize when this requirement can be passed to the right (64c)-(64d). Third, the elements that only have enclitic forms, like the *do* of British English *do*-ellipsis (Haddican, 2007), can encliticize when this requirement can be passed to auxiliaries to the left (64e)-(64f).

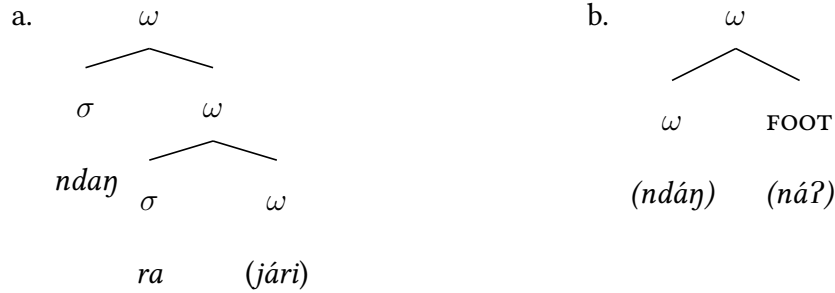
(64) *The Phonological ECP: English*

- a. I'd never go, but [_ω you] [_ω should / could / would / will / have / are]
- b. [_ω [_ω you] *sh' *c' *'d **ll **ve **re]
- c. ??I wouldn't be seen, but [_ω they'd] [_ω be]
- d. ??I haven't been seen, but [_ω they've] [_ω been]
- e. I wouldn't be seen, but [_ω they] [_ω would **a**]
- f. I wouldn't be seen, but [_ω they] [_ω might **do**]

In Mandar, I take these effects to suggest that the rightmost auxiliary must form a word (66a) and that above *na* this requirement is passed to the left (66b). In Section 5, we

will return to the pressures that operate in this context to force *na* to form a foot.

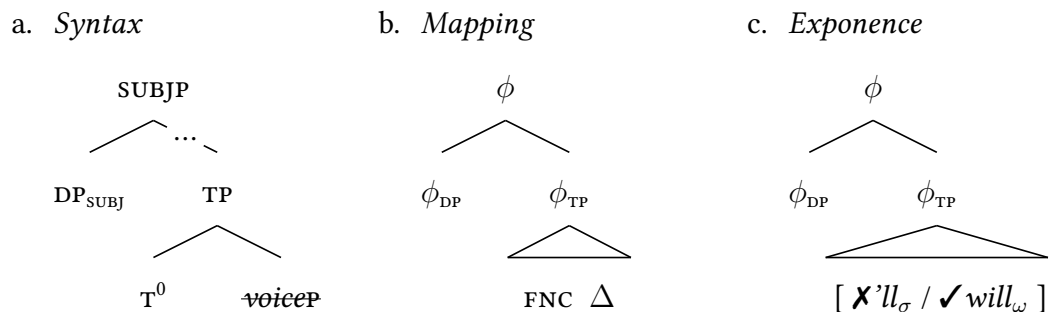
(65) *The Phonology of Stranding*



4.4 Recursive Prosodic Phrasing

To capture the Phonological ECP in English, [Anderson 2008a](#) proposes a two-part account (and see [Itô & Mester 2019](#) and [Kandybowicz 2015](#) for extensions). First, he proposes, the TP must (exclude the subject and) correspond to a phonological phrase ([66a](#)), yielding the representation in ([66b](#)) when auxiliaries are stranded by VPE. Second, he argues, every phonological phrase must eventually dominate a prosodic word—tracing the classic principle of HEADEDNESS ([Nespor & Vogel 1986](#); [Selkirk 1995](#)). In the default case, on Anderson’s account, this restriction is satisfied by the prosodic word built around the lexical head of the VP. But in contexts of ellipsis, functional heads are forced to satisfy it alone—ruling out enclitics and forcing the promotion in ([66c](#)).

(66) *Anderson’s Account*



In Mandarin, the distribution of the high tone reveals that this account is correct. Before the sites of Polarity Ellipsis and vPE, the final auxiliary always hosts a high tone—revealing that phonological phrases are always built around the POLP (67a) and TP (67b).

(67) *Phonological Phrasing before Ellipsis Sites*

- a. usáŋe^H jari lúlus^H na **jári^H** .
u-saŋa i iari lulus na iari Δ
1ERG-think 3ABS COMPLETE graduate and COMPLETE
‘I thought she graduated and [she] DID.’
- b. itʃittʃi^{HH} lúlus^H, tapi ikátʃo[?]^H ndar ra **jári^H** **áʔ^H**
itʃittʃi[?] lulus tapi ikátʃo[?] ndarŋ rua iari Δ a
NAME graduate but NAME not ever COMPLETE PRT
‘CHICHI’ graduated, but KACHO’ didn’t, eh.’

This generalization also extends to the complete set of environments where functional heads are stranded by movement. For instance, Mandarin allows prepositions to be stranded by various types of movement, and in this context, they show the same shift: they begin to form prosodic words and phonological phrases (68a)-(68b). The same effect extends to auxiliaries stranded by VP-topicalization (68c).

(68) *Preposition Stranding*

- a. wánne^H **tama** .i kóta[?]^H b. kóta[?] ínna^H mánne^H **táma^H** ?
u-anna i tama di kóta[?] kóta[?] inna mu-annai tama ____
1ERG-put 3ABS into P box box which 2ERG-put into
‘I put it into the box.’ ‘Which box did you put it into?’
- c. so ri jáwa^H ikátʃo[?] **bjása^H** i .
sau di dʒawa ikátʃo[?] biasa i ____
[voice go P⁰ PLACE] NAME usually 3ABS
- ‘Go to Java, Kacho does often.’

It also extends to the configurations where demonstratives and prepositions surface without overt complements in the extended NP. In these contexts, they form prosodic

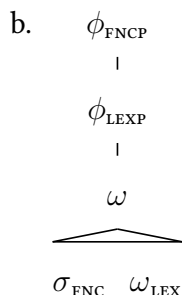
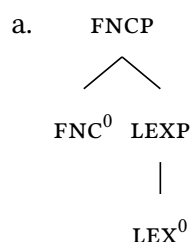
words—as in English—and host tones that mark the edges of phonological phrases built around the DEMP and PP (69a)-(69b). They contrast in this way with the language-internal class of weak pronouns (Cardinaletti & Starke, 1999), which never form prosodic words (as in English: Selkirk 1995) and never form phonological phrases (69c)-(69d). I thus submit that these demonstratives and prepositions are stranded by ellipsis as well.

(69) *Extending the Paradigm*

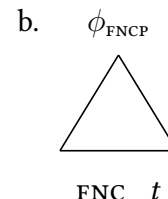
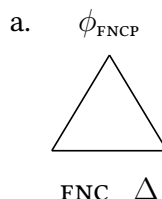
- | | |
|---|---|
| <p>a. əláŋã^H ítɪŋ^H áʔ^H.
 alaŋaŋ aʔ itɪŋ Δ a
 get for 1ABS that PRT
 ‘Get that for me, eh.’</p> | <p>b. áŋne^H náwuŋ^H áʔ^H.
 anna i nauŋ Δ a
 put 3ABS down PRT
 ‘Put it down, eh.’</p> |
| <p>c. póle^H ta kápaŋ^H.
 pole i tia kapaŋ
 come 3ABS 3SG maybe
 ‘Maybe she came’</p> | <p>d. póle^H tu kápaŋ^H.
 pole i tu kapaŋ
 come 3ABS IT maybe
 ‘Maybe it came’</p> |

To capture these effects, I submit that phonological phrases are preferentially built around every XP of the syntax. When functional heads surface before overt complements, this preference forces phonological phrases to be built around both functional and lexical xps (70). When lower xps are suppressed by movement or ellipsis, however, these higher layers of phonological phrasing become visible at the surface (71). The result is a straightforward argument for the central intuition of Match Theory: phonological phrases are preferentially built around every XP (Selkirk 2009, 2011).

(70) *Mapping*



(71) *Stranding*



5 Output Optimization

We can round out our investigation with a turn to the final phonological effects that emerge around the system of ellipsis. These are patterns of movement and allomorph selection that trace two top-down constraints (Booij, 1999; Prieto, 2005; Elordieta, 2006): a requirement for disyllabic feet at the right edge of the phonological phrase and a requirement for monosyllabic feet at the right edge of the intonational phrase. I take these effects to show that output constraints can shape the final stages of allomorph selection and linearization and then link them to the wider theory of the PF branch.

5.1 The Phonological Conspiracy

In phrase-medial positions, the future proclitic and the negator take their usual positions and forms: when they fall before auxiliaries and verbs (72a)-(72b) and when the negator falls before *bandi* (72c), which requires its host to form a prosodic word.

(72) Phrase-Medial Positions

- | | | |
|--|---|---|
| a. na balíli ^H i.
na balili i
FUT return 3ABS
‘She’ll come back.’ | b. íta [?] ^H ndam mála ^H .
ita [?] ndaŋ mala
2SG not can
‘YOU can’t.’ | c. ndám bàndá [?] ^H .
ndaŋ banda [?]
not VERUM.1ABS
‘I can’t?’ |
|--|---|---|

The future proclitic and the negator also retain their usual positions and shapes at the right edges of phonological phrases that are final in the intonational phrase. When the future proclitic is final in this environment, it shows epenthesis of a final glottal stop.

(73) Utterance-Final Positions

- | | |
|--|--|
| a. ikátʃo [?] ^H ndán ^H .
ikátʃo [?] ndaŋ Δ
NAME not
‘KACHO’ didn’t.’ | b. ikátʃo [?] ^H ndàn ná [?] ^H .
ikátʃo [?] ndaŋ na Δ
NAME not FUT
‘KACHO’ won’t.’ |
|--|--|

At the right edges of all other phonological phrases, they react in a conspiratorial way. The negator takes the form *andiang* (phonetically [əndiaŋ]) before phrase-external elements of all types, like final particles (74a) and absolutive and aspectual clitics (74b).

(74) *Other Phrase-Final Positions: Suppletion*

- | | |
|--|---|
| <p>a. ikátʃoʔ^H əndíáŋ^H áʔ^H.
 ikátʃoʔ ndaŋ Δ a
 NAME not
 ‘KACHO’ didn’t, eh?’</p> | <p>b. ikátʃoʔ^H əndíáŋ^H i .
 ikátʃoʔ ndaŋ i Δ
 NAME not 3ABS
 ‘Kacho’ didn’t.’</p> |
|--|---|

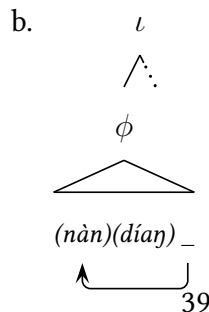
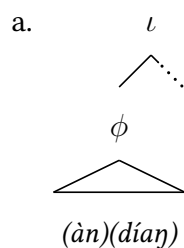
In the same contexts, the future proclitic shifts left. In some cases, this process only shifts it across the negator (75a), and in others, it carries it across second-position that should follow that element as well (75b). In each case, the /*na-andiang*/ string undergoes a regular process of coalescence and the two elements form a minimal prosodic word.

(75) *Other Phrase-Final Positions: Movement*

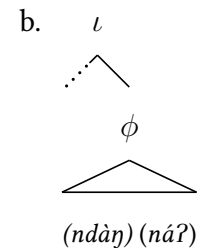
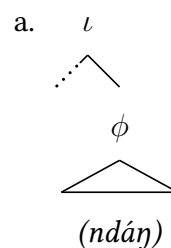
- | | |
|--|--|
| <p>a. ikátʃoʔ^H nàndíáŋ^H — áʔ^H.
 ikátʃoʔ na ndaŋ Δ a
 NAME FUT not
 ‘KACHO’ didn’t, eh?’</p> | <p>b. ikátʃoʔ^H nàndíáŋ^H i — .
 ikátʃoʔ na ndaŋ i Δ
 NAME FUT not 3ABS
 ‘Kacho’ won’t.’</p> |
|--|--|

The result is a system where two repairs conspire to achieve a consistent prosodic result: suppletion and movement both eliminate phrase-final monosyllabic elements (76) and both processes are suspended at the right edge of the intonational phrase (77).

(76) *Conspiracy*



(77) *Suspension*

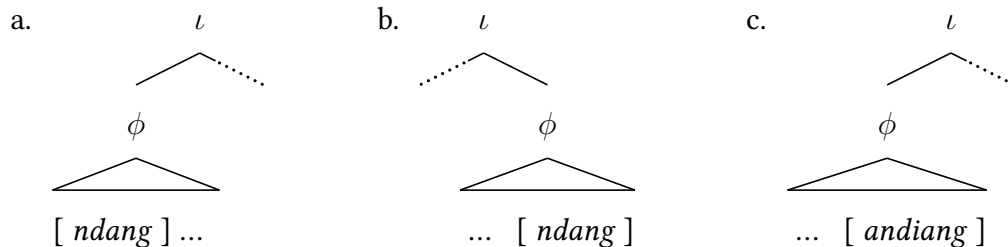


5.2 Identifying Suppletion

The *ndang/andiang* alternation is suppletive: these forms are listed allomorphs of Σ^0 that are selected in different environments by the system of exponence. This is the type of analysis that is universally upheld for English alternations like *would*–*'d* (Kaisse 1983, 1985; Wilder 1997; Anderson 2008a; Itô & Mester 2019; for fine empirical argumentation, see Inkelas & Zec 1993; Close 2004 and MacKenzie 2011, 2012, 2013). It contrasts with an alternative, advanced in English by Selkirk 1984, which would derive forms like *ndang* from forms like *andiang* by item-specific processes in the synchronic phonology.

The case for suppletion proceeds in three parts. The first is an argument from categoricity. When speakers are asked to reflect on the distribution of short and long forms in contexts where the distribution of focus is controlled, they are able to provide the categorical judgment in (78): *ndang* appears phrase-medially and in phrase-final positions that are final in the utterance and *andiang* appears in phrase-final positions that are not. In regular speech, this rule is only derailed by the phonology of focus and the long arm of production planning, which drive many deviations from the unmarked phonology of well-planned speech (Wagner 2011; MacKenzie 2013, 2016).

(78) Categorical Distribution



The second is an argument from gradience. In English, many auxiliaries show the paradigms like (79): their long forms have counterparts with vowel reduction and their short forms have counterparts with vowel epenthesis (Kaisse 1985; Inkelas & Zec 1993;

MacKenzie 2013). The Mandar forms show a similar effect: it is possible to reduce the initial /a/ in *andiang* to [ə], tracing the regular phonology of in the language, and it is possible to epenthesize [ə] before the short *ndang* and reduce its initial nasal in some corresponding way, yielding the alternation [*ndan̩* / əⁿ*dan̩*]. But there are no forms that are phonologically intermediate between the two—like a version of the long form with initial apocope (*ndiang*) or a version of the short form with the initial /a/ (*andang*). This fact suggests that they are not linked by regular processes of the synchronic phonology.

(79) *Missing Intermediate Forms*

LONG		INTERMEDIATE		SHORT	
<i>wʊd</i>	<i>wəd</i>	<i>✗wd</i>		<i>əd</i>	<i>d</i>
<i>wɪl</i>	<i>wəl</i>	<i>✗wl</i>		<i>əl</i>	<i>l</i>
<i>hæv</i>	<i>həv</i>	<i>✗hv</i>		<i>əv</i>	<i>v</i>
<i>andian̩</i>	<i>əndan̩</i>	<i>✗ndan̩</i>	<i>✗andan̩</i>	<i>əⁿdan̩</i>	<i>ndan̩</i>

The third is a two-part case from exceptionality. There is no attested process in the language that deletes initial /a/ or [ə]. Outside contexts of “pronunciation focus,” where the segmental shape of a word is deliberately stressed, the language preferentially reduces initial vowels to [ə] outside the final two syllables of the phonological phrase. This process generally targets the initial segments of /*andiang*/ and segmentally similar words in phrase-medial and utterance-final contexts (80a)-(80b). But in roots of that shape and in *andiang* it is impossible in all positions for initial vowels to delete.

(80) *No Initial Apocope*

a. *ənd̪ʒoro* *ɲura*^H
and̪ʒoro *ɲura*
 coconut young
 ‘Fresh coconut.’

b. *malli* *ənd̪ʒoro*^H *áʔ*^H
malli *i* *and̪ʒoro* *a*
 buy 3ABS coconut PRT
 ‘They bought coconuts, eh.’

At the right edge of the intonational phrase, in turn, the language prohibits deletion of prevocalic /i u/. In phrase-medial positions, these vowels typically delete between consonants and /i e o u/ (81a)-(81b). Before /a/, however, they cannot delete in the final foot of the word: there, /ia/ reduces to [e] and /ua/ reduces to [o] (81c)-(81d).

(81) *High Vowel Deletion*

- | | |
|---|--|
| <p>a. búk kàjján^H.
 biʔuŋ kaiaŋ
 rake big
 ‘A big rake.’</p> | <p>b. kíʔ létte^H i.
 kuiʔ letteʔ i
 remove by foot 3ABS
 ‘Kick it away.’</p> |
| <p>c. páre mapía^H.
 paria mapia
 gourd good
 ‘Good bitter gourd.’</p> | <p>d. báno kàjján^H.
 banua kaiaŋ
 hall big
 ‘Great hall.’</p> |

At the right edge of the intonational phrase, these processes are blocked and all vv sequences are retained. In the auxiliaries *biasa* and *rua*, the sequences /ia/ and /ua/ thus coalesce phrase-medially (82a)-(82b) but remain intact utterance-finally (82c)-(82d).

(82) *No Vowel Deletion*

- | | |
|---|---|
| <p>a. bésa wàndáʔ^H .
 biasa bandaʔ Δ
 usually VERUM.1ABS
 ‘I usually do.’</p> | <p>b. ró wàndáʔ^H .
 rua bandaʔ Δ
 once VERUM.1ABS
 ‘I once did.’</p> |
| <p>c. tapi bjása^H .
 tapi biasa Δ
 but usually
 ‘But [they] usually do!’</p> | <p>d. tapi ruwa^H .
 tapi rua Δ
 but once
 ‘But [they] once did!’</p> |

I take these facts to show that the regular phonology should categorically prohibit the reduction of a hypothetical form like **andang* to *ndang* and specifically prohibit the reduction of a hypothetical **ndiang* to *ndang* at the right edge of the intonational phrase.

5.3 The Top-Down Constraints

In Mandar, the right edge of the phonological phrase imposes a top-down requirement for *Positional Binariness*: the final foot must be disyllabic. This constraint drives two repairs in roots that typically end in monosyllabic feet: in roots of the shape cv(c), it forces a type of ?v-epenthesis (83a)-(83b), and in roots of the shape cvccvc, it drives various types of segmental deletion to allow the two syllables to be parsed into a moraic trochee. In roots of the shape cvccv?, for instance, final glottals delete (83c)-(83d).

(83) Positional Branching

- | | |
|---|--|
| <p>a. əláŋãʔ^H tʃáʔ maméa^H.
 alaŋaŋ aʔ tʃaʔ mamea
 get for 1ABS paint red
 ‘Get me some red paint.’</p> | <p>b. əláŋãʔ^H tʃáʔaʔ^H áʔ^H.
 alaŋaŋ aʔ tʃaʔ a
 get for 1ABS paint PRT
 ‘Get me some paint, eh.’</p> |
| <p>c. əláŋãʔ^H bállóʔ máŋaŋ^H.
 alaŋaŋ aʔ balloʔ maŋaŋ
 get for 1ABS wine strong
 ‘Get me strong palm wine.’</p> | <p>d. əláŋãʔ^H bálla^H áʔ^H.
 alaŋaŋ aʔ balloʔ a
 get for 1ABS wine PRT
 ‘Get me palm wine, eh.’</p> |

This pressure forces the same repairs onto stranded functional heads: at the right edge of the phonological phrase, ?v-epenthesis applies to cv-heads like *da* “don’t” and *sung* “out” (84a)-(84b) and ?-deletion applies to the cvccvc auxiliary *lesse*’ (84c)-(84d).

(84) Repairing Stranded Heads

- | | |
|---|---|
| <p>a. do wái^H ɿa muúndu^H óʔ^H.
 do uai da mu-dundu o
 that water don’t 2ERG-drink PRT
 ‘Don’t drink that water there.’</p> | <p>b. do wái^H ɿáʔa^H óʔ^H.
 do uai da Δ o
 that water don’t PRT
 ‘That water there, don’t.’</p> |
| <p>c. ikátʃoʔ^H lesseʔ láo^H áʔ^H.
 ikátʃoʔ lesseʔ lao a
 NAME INCOMPLETE go PRT
 ‘Kacho’ has started to go, eh?’</p> | <p>d. ikátʃoʔ^H lésse^H áʔ^H.
 ikátʃoʔ lesseʔ Δ a
 NAME INCOMPLETE PRT
 ‘Kacho’ has started to, eh?’</p> |

At the right edge of the intonational phrase, this constraint is not enforced. In phrase-final positions at that edge, then, cv(c) roots cease to show ?v-epenthesis (85a)-(85b) and cvccv? roots cease to show glottal deletion (85c)-(85d).

(85) *The Suspension Effect*

- | | |
|---|--|
| <p>a. əláŋãʔ^H tʃáʔ^H.
 aləŋaŋ aʔ tʃaʔ
 get for 1ABS paint
 ‘Get me some paint.’</p> | <p>b. itĩ wái^H ɬáʔ^H .
 itiŋ uai da Δ
 that water don’t
 ‘That water, don’t.’</p> |
| <p>c. əláŋãʔ^H bàllóʔ^H.
 aləŋaŋ aʔ balloʔ
 get for 1ABS palm wine
 ‘Get me some palm wine.’</p> | <p>d. ikátʃoʔ^H lèsséʔ^H .
 ikatʃoʔ lesseʔ Δ
 NAME INCOMPLETE
 ‘Kacho’ has started to’</p> |

This shift is forced by a second and competing top-down constraint: in Mandar, the intonational phrase preferentially ends in a monosyllabic foot. This effect can be seen from another part of the system of final ?-epenthesis: this process targets roots that are exhaustively parsed without it, like the words in (86a)-(86b), whenever it can yield final monosyllabic feet without derailing the parse of the rest of the word (86c)-(86d).

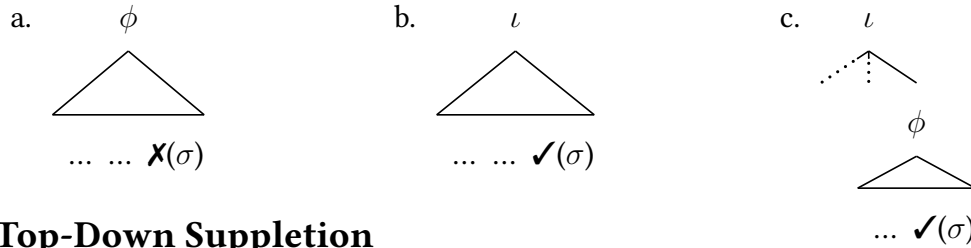
(86) *Building Monosyllabic Feet*

- | | |
|---|--|
| <p>a. əláŋãʔ^H əndʒóro ɲúra^H.
 aləŋaŋ aʔ andʒoro ɲura
 get for 1ABS coconut young
 ‘Get me some young coconut.’</p> | <p>b. pìssáŋɲi^H əkkátta wáru^H.
 pissanɲi i akkatta baru
 tell 3ABS plan new
 ‘Tell him the new plan.’</p> |
| <p>c. əláŋãʔ^H əndʒoróʔ^H.
 aləŋaŋ aʔ andʒoro
 get for 1ABS coconut
 ‘Get me some coconut.’</p> | <p>d. pìssáŋɲi^H əkkàttáʔ^H.
 pissanɲi i akkatta
 tell 3ABS plan
 ‘Tell him the plan.’</p> |

The following diagrams summarize this result: the phonological phrase demands a final disyllabic foot (87a), the intonational phrase demands a final monosyllabic foot

(87b), and when these pressures conflict, the intonational phrase wins out (87c). This is an interaction between two types of top-down constraints that is relatively familiar at many other levels of the prosodic hierarchy (Booij, 1999; Prieto, 2005; Elordieta, 2006).

(87) *The System of Minimality*



5.4 Top-Down Suppletion

When the negator is shielded from the Positional Binarity Constraint, in phrase-medial and intonational phrase-final positions, it takes the form *ndang* (88a)-(88b). When exposed to that constraint, it suppletes to *andiang* (88c)-(88d). This is a type of allomorph selection that is sensitive to top-down prosodic context, on a par with the effects described by Tucker & Henderson 2010; Henderson 2012 and McCarthy 2012, and one that is driven by an output constraint (Mester, 1994; Mascaró, 1996b; Perlmutter, 1998).

(88) *The Suspension Effect*

a. ndám bándi^H làmbáʔ^H? ndaŋ bandi lamba not VERUM go ‘Did he not go?’	b. tàniáʔ^H, ndán^H tania ndaŋ Δ not so not ‘No, [he] didn’t.’
c. tàniáʔ^H, andían^H i tania ndaŋ i Δ not so not 3ABS ‘No, he didn’t.’	d. ikátʃoʔ^H andían^H áʔ^H? ikatʃoʔ ndaŋ Δ a NAME not PRT ‘KACHO’ didn’t.’

I propose that output-optimizing allomorph selection emerges when four conditions are met. First, the morphology must leave the phonology with a choice (Tranel, 1996;

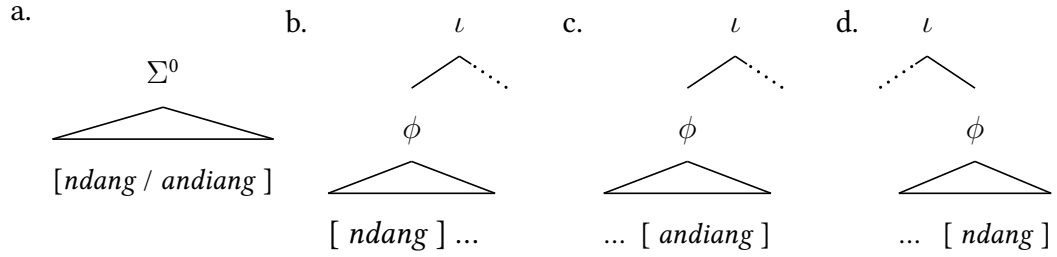
Rubach & Booij, 2001; Klein, 2003): the output of MATCH must be a set that contains multiple possible exponents. Second, individual allomorphs in that set must be favored by output constraints, like the Positional Binarity Constraint (89a) and the Positional Alignment Constraint (89b). Third, those constraints must outrank the pressure to insert the allomorphs that are lexically unmarked (89c), enforced by the constraint PRIORITY (Mascaró 1996a,b, 2007; for an alternative approach, see Kager 1996). Fourth, those constraints must also outrank the pressure to satisfy the subcategorization frames of individual allomorphs, which are otherwise able to drive patterns of phonologically conditioned allomorph selection that cut against output constraints (Bonet *et al.* 2007; for discussion, see also Booij 1998, 2012; Paster 2006; Bye 2007).

(89) *Licensing Optimization*

- a. POSITIONAL BINARITY: AOV for every ϕ that does not end in a branching foot.
- b. POSITIONAL ALIGNMENT: AOV for every ι that does not end in a unary foot.
- c. PRIORITY: Given an input containing an x^0 that can be realized with the lexically ordered allomorphs $\{M_1, M_2 \dots M_n\}$ and a candidate that contains M_i , assign as many violation marks as the depth of ordering between M_i and M_1 .

This framework allows us to capture the *ndang/andiang* alternation in five steps. First, I submit that both of these allomorphs must be selected in all contexts as possible exponents for Σ^0 in the morphology (90a). Second, I assume that neither is associated with a phonological subcategorization frame, ruling out interference from SUBCAT. Third, I assume that *ndang* is lexically listed as the unmarked exponent of Σ^0 . As a result, PRIORITY forces insertion of *ndang* in phrase-medial contexts like (90b). Fourth, I propose that the Positional Binarity Constraint outranks PRIORITY and forces insertion of *andiang* in the phrase-final context in (90c). Fifth, I propose that the Positional Alignment Constraint forces insertion of *ndang* in the utterance-final context in (90d).

(90) *Deriving the Alternation*



5.5 Phonological Relinearization

The same constraint forces the future proclitic to shift. When *na* is shielded from the Positional Binariness Constraint, it remains in place (91a). At the edge of the intonational phrase, in turn, it undergoes final glottal epenthesis to host a monosyllabic foot (91b).

(91) *Staying in Place*

- a. $\widehat{\text{ikátfo}}^{\text{H}} \text{ndan } \mathbf{na} \text{ } \mathbf{mála}^{\text{H}}$.
 $\widehat{\text{ikátfo}} \text{ndan} \text{na} \text{mala} \Delta$
 NAME not FUT can
 ‘KACHO’ won’t be able to.’
- b. $\widehat{\text{ikátfo}}^{\text{H}} \mathbf{ndàn} \mathbf{ná}^{\text{H}}?$.
 $\widehat{\text{ikátfo}} \text{ndan} \text{na} \Delta$
 NAME not FUT
 ‘KACHO’ won’t.’

When *na* is exposed to the Positional Binariness Constraint, it moves: shifting across the negator in some cases (92a) and crossing second-position clitics in others (92b).

(92) *Time Travel*

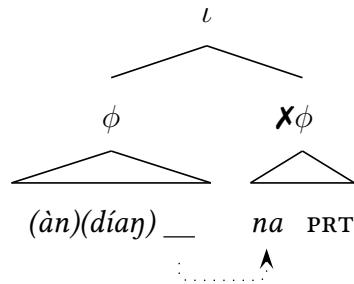
- a. $\widehat{\text{ikátfo}}^{\text{H}} \mathbf{nàndián}^{\text{H}} \text{ } \text{á?}^{\text{H}}?$
 $\widehat{\text{ikátfo}} \text{na} \text{ndan} \Delta \text{a}$
 NAME FUT not PRT
 ‘Kacho’ won’t, eh.’
- b. $\widehat{\text{ikátfo}}^{\text{H}} \mathbf{nàndián}^{\text{H}} \text{ i } \text{ } \text{ } ?$
 $\widehat{\text{ikátfo}} \text{na} \text{ndan} \text{i} \Delta$
 NAME FUT not 3ABS
 ‘Kacho’ won’t.’

I propose that the Positional Binariness Constraint forces this step of movement through the following interaction of constraints. When *na* falls at the right edge of the phonological phrase before a final particle, it cannot be parsed into a word with that

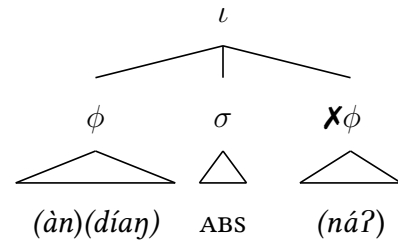
element (93a) because of DEP- ω (53a), which favors the construction of words around heads in a single extended projection (whatever the exact syntax of these particles may be). When it falls at the right edge of a phrase after the second-position clitics, in turn, it cannot form a prosodic word (93b) because of its prosodic subcategorization frame (50a).

(93) *Impossible Alternatives*

a. *Rebracketing*



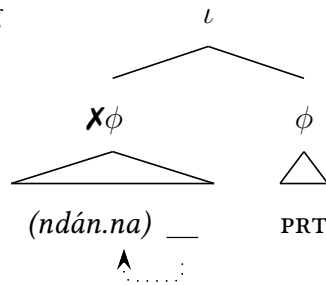
b. *Independence*



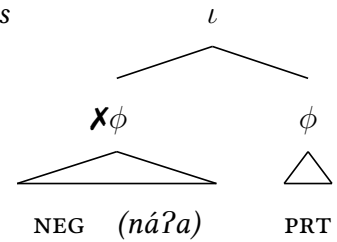
Na is then prevented from satisfying Positional Binarity Constraint in-situ by two item-specific constraints (on which see Pater 2000, 2006, 2010 and Gouskova 2012): one which prevents it from falling into the weak branch of a foot with the negator (94a) and one that prevents it from undergoing the regular repair of ʔv -epenthesis (94b).

(94) *Impossible Repairs*

a. *Refooting*



b. *Epenthesis*



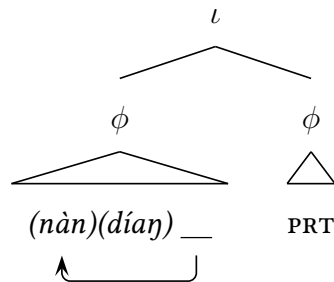
5.6 The Theory of Linearization

Following the standard theory of linearization in DM, I assume that linear order is absent from the narrow syntax and initially established in the morphology (Embick &

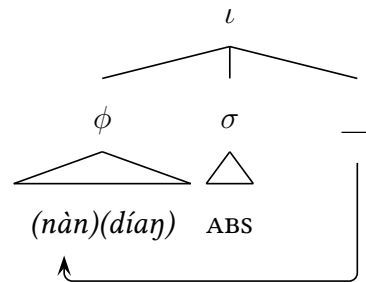
Noyer 2001; Embick 2007; see also Chomsky 1995). In that module, I take it to be established by a dedicated system of linearization that precedes MATCH, allowing it to feed morphological allomorph selection (Adger, 2006; Merchant & Pavlou, 2017; Ostrove, 2018; Hewett, 2023) and various processes keyed to the edges of morphological domains (Arregi & Nevins 2012). After this stage, I assume that it can be modified by morphological operations like Local Dislocation (Embick & Noyer, 2001) or the Generalized Reduplication and reduction of Harris & Halle 2005 (Arregi & Nevins, 2012; Hewett, 2023). In the phonology, I propose that it can then be reworked in service of output constraints. To capture the facts of Time Travel, concretely, I propose that the exponent *na* is shifted to the left as a repair to the Positional Binararity Constraint (95).

(95) *Phonological Displacement*

a. *Short Distance*



b. *Long Distance*

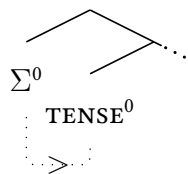


This analysis treats Time Travel as a type of PUSH-movement, in the sense of (Diesing, 1992; Moro, 2000; Stroik, 2009; Chomsky, 2013), that is driven by an output phonological constraint (Inkelas & Zec, 1990; Halpern, 1992, 1995; Schütze, 1994a,b; King, 1996; Inkelas & Zec, 1990; Zec, 2005; Diesing & Zec, 2011, 2017; Bennett *et al.*, 2016). It contrasts with an alternative that would draw *na* through a series of positions in the morphosyntax and simply allow the phonology to select one of them for realization, on a par with the analysis of second-position effects in Bošković 2001. On this type of account, we might understand *na* to consistently undergo the morphological process of Local Dislocation in (96a)-(96b). In the usual case, it would be spelled out in

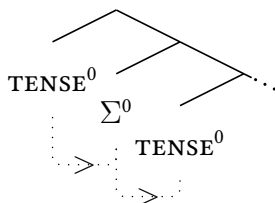
its original position (96c), but when that result would violate the Positional Binariness Constraint, it would then be spelled out in the derived position in (96d).

(96) *Alternative Analysis: Chain Reduction*

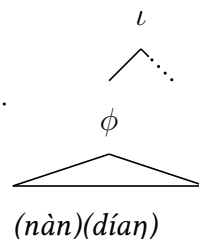
a. *Linearization*



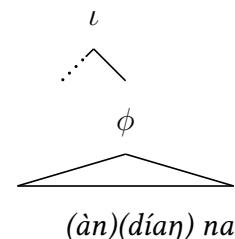
b. *Dislocation*



c. *ι-Final*



d. *ι-Medial*



To capture the facts of Time Travel, this analysis seems plausible and interesting, as it would require the system of chain reduction (which may be the system of exponence) to interact with output constraints. My argument against it is purely conceptual. The literature has identified a range of cases where elements are unambiguously positioned in the phonology, in response to output constraints (Noyer, 1994; Hargus & Tuttle, 1997; Kim, 2010; Zukoff, 2020) or subcategorizational frames (Radanović-Kocić, 1988; Anderson, 1996, 2008b; Chung, 2003; Aissen, 2017; Ionova, 2019; Weisser, 2020; Belkind, 2023; Aremu *et al.*, 2024). In Mandarin, for instance, Brodsky 2026 shows the following effect: the language has several demonstratives that lexically select reinforcers (97a), these elements form phonological phrases and typically host degenerate feet (97b), and whenever they are initially linearized into positions that are non-final in the intonational phrase, they shift to the right: escaping coordinate structures (97c) and embedded clauses (97d) and reacting to prosodic shifts forced by metrical line breaks (97e)-(97f).

(97) *Phonological Displacement*

- a. **Ndi** posa ^{*}(e), **de** posa ^{*}(e), **do** posa ^{*}(o), iting posa (o).
 this cat here this cat here that cat there that cat there
 ‘This cat right here, this cat here, that cat there, that cat over there.’

- b. {_φ **do** pósa }^H {_φ **óʔ** }^H
do posa o
that cat there
‘That cat there’
- c. U-ita i [_{CP} do tau ____ na sola-na iKaco’] allo ajuma’ **o**.
1ERG-see 3ABS that guy and friend-3GEN NAME friday there
‘I saw that guy there and Kacho’s friend on Friday.’
- d. Ma’ita a’ [_{DP} karewa [_{CP} mua’ pole i do tau ____]] dio di FB **o**.
I saw news that come 3ABS that guy there on FB there
‘I saw [_{DP} news [_{CP} that that guy there is coming]] on facebook.’
- e. {_ℓ jamo **io** ____ .isaŋa lopi pattonda ropponj **oʔ** }
jamo do disaŋa lopi pattonda ropponj **o** }
is that called boat shipping grass there
‘That there is called a worthless endeavor.’
- f. {_ℓ jamo **io** ____ .isaŋa **oʔ** } {_ℓ lopi pattonda ropponj }
jamo do disaŋa **o** } {_ℓ lopi pattonda ropponj }
is that called there boat shipping grass
‘{_ℓ That there is called } {_ℓ a worthless endeavor }.’ Muthalib & Sangi 1991: 374

To allow phonological displacement of this type, it seems reasonable to imagine that the linearization of the morphology is reinstantiated over the output correspondents of its terminal nodes by a ranked and violable interface constraint (McCarthy & Prince, 1993; Halpern, 1995). I take this to be the constraint in (98), which checks the linear order of terminal nodes in the morphology against the order of their exponents in the phonology (see also Göbbel 2007; Werle 2009; Bennett *et al.* 2016; Aissen 2017; Clemens & Coon 2018; Clemens 2019; Kusmer 2020; Potsdam 2022). This approach allows us to understand phonological displacement as a mismatch between the linearization of the phonology and the linearization of the morphosyntax: a type of imperfect translation that emerges in the ranking configurations that McCarthy & Prince 1993 term P>M (and see Selkirk 2009, 2011 for a theory of prosodic structure-building that works in this way).

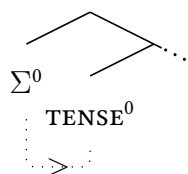
(98) NO SHIFT

Assign one violation for every pair of constituents α, β , such that α is linearized before β in the morphological input, for which the output correspondent of α is not linearized before the output correspondent of β in the phonological output.

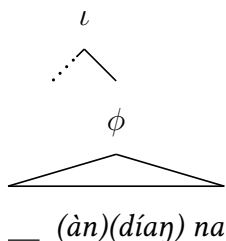
Against this backdrop, I submit the following argument from parsimony. It is possible to state the complete distribution of *na* in output-optimizing terms: it is linearized after the negator in the morphology (99a), it is realized in that position when it can escape the Positional Branching Constraint (99b), and when it cannot, *na* moves the minimum distance away from the right edge of the phonological phrase (99c). As there is no evidence here for morphological displacement, I propose that this account is correct and that *na* is relinearized in the phonology to satisfy an output constraint.

(99) *Phonological Movement: Revised*

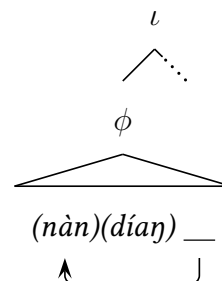
a. *Linearization*



b. *Staying in Place*



c. *Moving to the Left*



6 Conclusion

Drawing things together, we can summarize the analytical core of this work as follows. First, I have argued that there is a regular phonological constraint, the *Phonological ECP*, that emerges before the sites of movement and ellipsis and forces stranded functional heads to form prosodic words. It emerges from the interaction of two output constraints: the requirement to build prosodic words in phonological phrases and the preference to build phonological phrases around syntactic xps. This constraint yields the restrictions

on enclisis that emerge before VPE sites in English, it is distinct from the syntactic licensing conditions on ellipsis, and it links ellipsis with movement in a surface respect.

Second, I have argued that this restriction and its phonological consequences can interact with the listed properties of individual exponents to force three output-oriented repairs: movement, suppletion, and salvation of material in ellipsis sites. The first effect provides a convergent argument for relinearization in the phonology, while the second reinforces the case that output constraints can guide allomorph selection—providing further evidence especially for interactions with top-down prosodic constraints. The third result provides evidence for three points. First, there must be morphosyntactic structure inside of certain ellipsis sites. Second, that structure must be suppressed in the phonology in a manner distinct from phonological deletion. Third, that structure must also persist until the derivation reaches the phonology—suggesting that in the morphology the contents of ellipsis sites are simply marked for non-insertion.

We can conclude our work with the following note. The literature has amassed a range of observations to suggest that ellipsis can interact with certain systems in the morphosyntax. On the analysis developed here, it seems necessary to link these effects to the steps that lead up to non-insertion. The cases where ellipsis seems to bleed phrasal movement in the syntax, for instance, might be bound up with the syntactic mechanisms beneath the assignment of [E] (on these cases see [Dikken *et al.* 2000](#); [Van Craenenbroeck & Den Dikken 2006](#); [Aelbrecht 2008, 2010, 2012](#)). The interactions between ellipsis and the operations of the morphology, in turn, might be bound up with morphological assignment of the feature [$\uparrow$] (and on these effects see [Saab & Lipták 2016](#); [Van Craenenbroeck & Temmerman 2017](#); [Murphy 2018](#)). The apparent interactions between ellipsis and phase-based spell-out, in turn, might be linked to be the systems of transfer from the syntax to the morphology (and on these effects see [Gengel 2005, 2013](#); [Gallego 2009](#); [Rouveret 2012](#); [Harwood 2013, 2015](#); [Bošković 2014](#); [Wurmbrand 2017](#)).

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Notes

¹The systems of interest in the present investigation—the future proclitic, the auxiliaries, and the processes of Polarity Ellipsis and *vPE*—seem to be shared by many languages of the South Sulawesi Subfamily, which have been described in a small body of work (Friborg, 1991, 1996; Strømme, 1994; Matti, 1994; Valkama, 1995a,b; Jukes, 2006; Lee, 2008; Kaufman, 2008; Laskowske, 2016; Finer, 1997, 1998, 1999; Béjar, 1999). Many of the languages closest to Mandar—in the Northern subgroup of the subfamily—also require metrical feet to branch at the right edge of the phonological phrase in the manner described in Section 5. In Mamuju, the closest relative of Mandar, this constraint also drives movement of the future proclitic before non-final *vPE* sites. The following example provides textual attestation of this effect.

(100) *Time Travel*

Mamuju: Rasyid & Usmar 1998, 6:46-50

- a. **De**=areki **na** diang tarento' ampunna' ingkai' domai molabu.
 not=PRT will there be waves if we come here to harbor
 'We hope there won't be any waves if we come here to harbor.'
- b. Ampunna' diang ki akkattata' na domai molabu,
 If there is your intention to come here to harbor
 'If you have the intention of coming here to harbor,
- tente mo inne: so'nai mo kipatipatindoang barang tallumbongi,
 like this leave it for us to sleep on roughly three nights
 'this is what we'll do: let us sleep on it for a few nights,
- ampele' mengkatanda a' ingkai' mating di ingkita'
 and then signal we to you
 'and then we will signal to you
- na** mabombang ai **na** **u'de** [vp — **mabombang**] ai.
 will turbulent PRT will not turbulent PRT
 'Whether it will be turbulent or whether it will not.'

²GLOSSING: ABS: absolutive, ANTIP: antipassive, ERG: ergative, GEN: genitive, PASS: passive.

³The same parse extends to the functional heads that are amalgamated with their hosts in the morphology (Harizanov & Gribanova, 2019; Arregi & Pietraszko, 2018, 2021, 2026), like *voice*⁰ and *v*⁰. The Mandar verb raises through these heads to pick up the voice prefixes in (101), and all of those elements fall outside the domain of stress and do not interact with glottal epenthesis.

(101) *Parallel: Prefixes*

- | | |
|---|--|
| <p>a. usáŋa^H pe ɲisáka^H.
 u-saŋa pai di-saka
 1ERG-think YET.3ABS PASS-catch
 ‘I think she’ll be caught.’</p> | <p>b. usáŋa^H pe tisáka^H.
 u-saŋa pai ti-saka
 1ERG-think YET.3ABS GET-catch
 ‘I think she’ll get caught.’</p> |
| <p>c. usáŋa^H po natùmaéʔ^H.
 u-saŋa pao na-tumae
 1ERG-think YET.2ABS 3ERG-ask
 ‘I think he’ll propose to you.’</p> | <p>d. usáŋa^H po siwàlelôʔ^H.
 u-saŋa pao si-balelo
 1ERG-think YET.2ABS WITH-fight
 ‘I think you guys will fight.’</p> |

The facts are the same with the adjectival prefix *ma-* and the propriial article *i* (102a)-(102b), though Mandar names raise to D^0 (see also Longobardi 1994) and adjectival roots may well raise to a^0 . It does not extend to suffixes, which reroute the systems of stress and glottal epenthesis in a surface-transparent way: for instance, possessive agreement in the DP (102c)-(102d) and applicative suffixes on the verb (102e)-(102f).

(102) *Stress Domain: Extensions*

- | | |
|--|---|
| <p>a. látto^H mi marása^H.
 latto mi ma-rasa
 looks NOW.3ABS ADJ-great
 ‘Now it looks great.’</p> | <p>b. pole^H pe iwòrahíma^H.
 pole pai i-borahima
 come YET.3ABS D-NAME
 ‘Abraham is coming yet.’</p> |
| <p>c. póleʔ^H pe solána áʔ^H?
 pole pai sola-na a
 come YET.3ABS pal-3GEN PRT
 ‘Their friend will come, eh?’</p> | <p>d. póleʔ^H pe sòlanáʔ^H.
 pole pai sola-na
 come YET.3ABS pal-3GEN
 ‘Their friend will come.’</p> |
| <p>e. usáŋa^H pe ɲipoléi^H déʔ^H.
 u-saŋa pai di-polei-i de
 1ERG-think YET.3ABS PASS-come-I PRT
 ‘I think she’ll be visited, I guess.’</p> | <p>f. usáŋa^H pe ɲipòleíʔ^H.
 u-saŋa pai di-polei-i
 1ERG-think YET.3ABS PASS-come-I
 ‘I think she’ll be visited.’</p> |